

# Null-Spinor Area as the Rest Gap of an Exactly Unitary Dirac Walk

## A Machine-Verified Finite Construction

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### Abstract

The rest gap is normally an independent input to a Dirac quantum walk. We derive its operator from supplied null-spinor geometry and prove, with a machine-checked finite development, exactly what that derivation adds. For  $P = \sum_i \psi_i \psi_i^\dagger$  built from massless rank-one momenta,

$$\det P = \sum_{i < j} |\psi_i \wedge \psi_j|^2,$$

and the invariant vanishes exactly when the occupied spinors span at most one complex direction. For two spinors, the complex area  $z = \psi_1 \wedge \psi_2$  uniquely determines the odd Hermitian rest operator

$$B_z = \begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}, \quad B_z^2 = \det(P)\mathbf{1}, \quad (k\sigma_z + B_z)^2 = (k^2 + \det P)\mathbf{1}.$$

Thus the scalar mass input is replaced by null-spinor data, not eliminated: the construction derives the form and geometric zero locus of the gap but does not predict an absolute mass scale.

Exponentiating  $B_z$  gives an exactly unitary null-step walk. Opposite turns carry  $z$  and  $\bar{z}$ , fixed-momentum symbols converge to Dirac flow uniformly on compact momentum sets at rate  $O(1/n)$ , and a position-dependent field  $z(x)$  yields an exactly reversible finite-torus update with a strict causal cone and explicit endpoint links under local chiral rotation. Most importantly, the oriented datum is already visible in the *free* one-particle spectrum. For equal-modulus site fields coupled by an edge phase  $w$ , the exact two-site Hamiltonian satisfies

$$(H^2 - (|z_L|^2 + t^2)\mathbf{1})^2 = t^2 |z_R - (\bar{w})^2 z_L|^2 \mathbf{1}.$$

Hence transported relative phase, unlike a common phase, cannot be reproduced by assigning only the scalar  $|z|$ . The same chiral covariance uniquely selects a phase-reading quartic pair coupling at block level; its local Fock placement, exact pulse law, and finite causal support are also verified.

The formal audit exposes a sharp regulator boundary rather than hiding it. In the ordered 3+1 successor, all zero- and  $\pi$ -quasienergy crossings are classified and their charges cancel at each gap, while exact no-go theorems show that the live minimal architecture retains high-frequency aliases and cannot open a global Floquet gap with a momentum-independent onsite coin. Accordingly, the result is a derived finite rest operator with a phase-sensitive free observable and a verified dynamical chain, not a completed continuum field theory. Position-space PDE convergence, an alias-free local 3 + 1 successor, a fully interleaved interacting Fock dynamics, and absolute mass selection remain open.

**How to read the status marks.** Origin and verification are independent. **CLASSICAL** identifies established mathematics formalized here; **NEW** identifies a new theorem or construction; **REALIZATION** identifies an explicit model. **KERNEL** means the stated finite claim is checked by the pinned Lean kernel; **PROOF** means a complete mathematical proof is given here but its dedicated Lean declaration is pending; **OPEN** marks a named unsolved problem. Physical interpretations are never promoted by a verification mark.

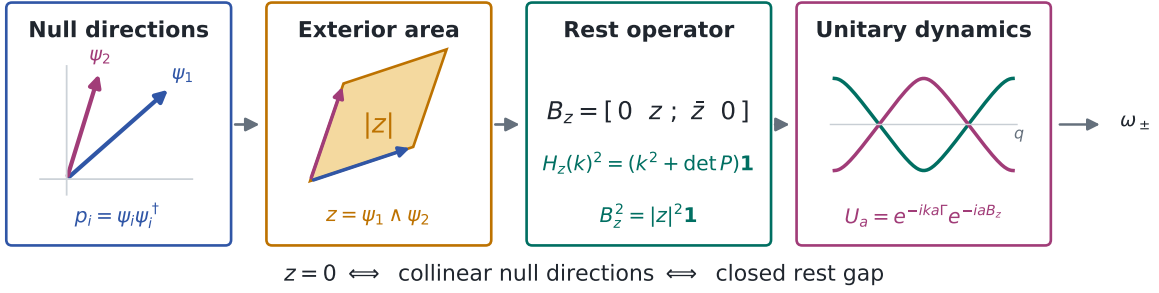


Figure 1: The complete finite construction. The input is a pair of null spinors, not a preassigned mass angle. The complex area fixes both the rest spectrum and the orientation phase of the unitary turn; collinearity is the exact gap-closing boundary.

## 1 Introduction: one invariant, one operator, one walk

The sum of lightlike momenta need not be lightlike. In the spinor representation, each future null momentum is a rank-one positive matrix  $\psi\psi^\dagger$ . A collection of such momenta remains on the null boundary exactly while its spinors occupy one complex direction. Once two directions open, their exterior area is nonzero and the sum moves into the timelike interior.

This familiar geometry contains a sharper operator statement. The complex area  $z = \psi \wedge \phi$  is not merely a number whose squared modulus happens to equal mass squared. It is the off-diagonal entry of a canonical odd Hermitian operator. That operator squares to the invariant momentum determinant, anticommutes with the velocity grading, and therefore supplies the rest term of a finite Dirac Hamiltonian. Its exponential is an exact unitary turn gate. The same object thus controls quadratic cost, first-order rest energy, finite-time rotation, and corner amplitudes.

**Synthesis theorem.** Let  $\psi, \phi \in \mathbb{C}^2$ , set  $P = \psi\psi^\dagger + \phi\phi^\dagger$ ,  $z = \psi \wedge \phi$ , and  $\mu = |z|$ . Then

$$\det P = \mu^2, \quad B_z = \begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}, \quad B_z^2 = \mu^2 \mathbf{1}.$$

Writing  $\Gamma = \text{diag}(1, -1)$  and  $H_z(k) = k\Gamma + B_z$ , one has

$$\Gamma B_z + B_z \Gamma = 0, \quad \boxed{H_z(k)^2 = (k^2 + \det P) \mathbf{1}}.$$

Thus the positive rest energy is  $\mu = \sqrt{\det P}$ , and the gap closes exactly on the collinear null boundary. Multiplying  $z$  by a unit phase changes  $B_z$  by unitary conjugation, so the spectrum depends only on  $P$  while the oriented decomposition retains a path phase. NEW

**KERNEL**

**What changes physically, and what does not.** The Pluecker origin fixes more than a dispersion parameter: it fixes the complex directed turn amplitudes, their local chiral connection under a spatially varying field, and the exact gap-closing locus. Transport makes the phase observable without adding an interaction: two equal-modulus rest fields can have different free spectra, with the splitting controlled exactly by  $\Delta = z_R - (\bar{w})^2 z_L$ . A covariance-selected quartic pair transfer provides a second, many-body phase reader. The construction does not by itself cure lattice aliases or predict an absolute mass. Within the four-channel separable degree-one architecture, every even-parity zone corner is provably locked to the origin after any onsite Pluecker coin; under the natural positive homogeneous action, every positive configuration has only the collapsed stationary scale. These are falsifiable structural consequences, not interpretive qualifications. NEW [KERNEL](#)

Three claims organize the paper. First, null-spinor area uniquely generates the rest operator and its finite dynamical chain. Second, transported relative phase is a free-spectrum observable rather than merely a supplied interaction phase. Third, the minimal ordered  $3 + 1$  architecture has a completely audited high-frequency boundary, turning the search for a clean strict regulator into a precise algebra problem. Seven technical contributions support those headlines.

1. **The invariant.** We isolate the exterior-area invariant, its exact massless boundary, and the grading between the first-order gap  $\mu$  and the quadratic invariant  $\mu^2$ . The Cauchy–Binet identity is classical; its complete finite formalization and controls are part of the contribution.
2. **The canonical operator.** We construct  $B_z$  directly from the complex Pluecker coordinate, prove the synthesis theorem for arbitrary spinor pairs, and classify every odd Hermitian rest operator as  $B_w$  for a unique  $w$ . No rational example or separately chosen Dirac mass parameter carries the general statement.
3. **Exact histories.** We derive the exact unitary turn gate and its corner expansion, with finite-spacing corner parameter  $\tan(a\mu)$  rather than  $a\mu$ . A directed-edge path theorem places  $z$  and  $\bar{z}$  in the same finite history sum and proves equality with the actual complex Pluecker transfer power.
4. **Uniform continuum control.** We prove kernel-checked, uniform bounded-momentum  $O(1/n)$  many-step estimates for the  $1 + 1$  and ordered  $3 + 1$  split-step symbols. For the explicit schedule  $n = K^4$ , the  $3 + 1$  estimate remains convergent on arbitrary parameter sequences satisfying  $B_4 \leq K$ , with an  $O(K^{-2}) \exp(|t|/K^3)$  envelope. The full complex phase is covered by unitary conjugacy and by the direct complex-coin composition, not discarded after taking its modulus.
5. **A finite local  $3 + 1$  walk with exact Fourier control.** We construct an exactly norm-preserving finite-torus walk with an explicit two-sided inverse and a strict one-shift-per-axis causal cone, whose ordered symbol is the product controlled by the  $3 + 1$  estimate. The finite-to-analytic momentum-sign bridge is exact, an iterated three-torus Plancherel theorem gives the exact  $N^{-3}$  wave-packet normalization, and a compact-support multiplier theorem yields momentum-space  $L^2$  convergence at every fixed time. Arbitrary position-dependent Pluecker fields evolve unitarily, and site-dependent chiral rotation induces exact endpoint links.
6. **The obstruction audit.** The same framework proves where it stops, and now does so with an exact all-zone criterion: the determinant factorizations  $\det(U^{(4)} \mp \mathbf{1}) = 4P_{0,\pi}$  hold at every real momentum and mass angle, so zero and  $\pi$  modes are exactly the zeros of two explicit polynomials — completely classified for  $0 < |\cos \theta| < 1$  as the simultaneous vanishing of all

three momentum cosines. The massless cubic regulator has three nonzero zero-quasienergy corner aliases; the massive body-center step retains exact  $+1$  and  $-1$  modes for every mass angle, so the regulator is globally ungapped; a degree-one nearest-neighbor factor admits no stationary amplitude compatible with origin normalization, exact unitarity, and the full Dirac tangent; and every momentum-independent onsite coin leaves all three even-parity aliases intact. Exact temporal blocking closes the mass subgroup but generates new ordered words in the full split family, and a homogeneous-action no-go proves that selecting a nonzero  $|z|$  requires an additional scale-bearing input.

7. **The many-body seed.** A generic occupation-basis construction satisfies all canonical anticommutation relations, and a four-mode quartic pair transfer has oriented two-particle amplitudes  $z$  and  $\bar{z}$ , exactly distinguished at the fixture  $z = 3 + 4i$ . The associated normalized pair kick is an explicit complex-linear automorphism, involutive and Fock-inner-product preserving; the Hermitian quartic operator equals the kick on the pair sector and the identity off it, and the kick fixes every one-particle basis state while acting nontrivially on the displayed two-particle state. An interference corollary shows that the supplied interaction sector retains the phase: equal-modulus fields with exactly conjugate one-particle rest operators are separated at the displayed fixtures  $z = 3 + 4i$  and  $z = 5$ , inside the supplied pair-kick sector, by a double-kick return probability of  $4/5$  versus  $1$ . This is not yet a second quantization of the spatial walk: the determinant-minor lift now has exact vacuum, one-particle, number, parity, creation-covariance, identity, and composition theorems, including  $\Gamma(UV) = \Gamma(U)\Gamma(V)$ , and its ordered-minor coefficients obey exact conjugate-transpose compatibility. A one-particle unitary's lifted conjugate transpose is an exact two-sided inverse, and the lift preserves the finite occupation-basis Hermitian inner product. Creation and annihilation covariance are exact. Separately, the pair kick can be placed on arbitrary four-mode sets; disjoint gates commute, and every finite declared schedule has exact support propagation in the even CAR algebra, with a nonzero phase-transfer witness inside the acted set. Under an explicit graph neighborhood and finite-range condition on every block, a sequential schedule obeys an exact iterated-neighborhood cone. Pairwise-disjoint local gates have now been grouped into layers, with one neighborhood expansion per layer and an exact radius bound by circuit depth. The free/interacting support composition is now exact in general form: determinant-minor second quantization of any coefficient-local finite-range unitary expands genuine CAR support by one graph neighborhood, and one disjoint local pair layer adds exactly one more. Instantiating this theorem with the complete live  $3 + 1$  walk, deriving an operator-exponential law, and computing scattering or binding remain open.

The result sits between spinor-helicity kinematics and discrete relativistic quantum dynamics [1, 2]. Cauchy–Binet and the Pauli map supply the classical foundation. Feynman's checkerboard and modern quantum walks show how a massive Dirac propagator can arise from null steps and direction changes [4, 5, 6, 7, 18]. Our contribution is the exact finite map from the oriented spinor area to the odd rest operator and then to a controlled unitary walk, together with a machine-checked dependency boundary.

## Nearest constructions and the exact novelty claim

For constant  $z$ , the update in this paper is unitarily equivalent to the standard  $1 + 1$  split-step Dirac walk. We do not claim otherwise. What is new is the input map and its naturality: the coin is generated by the complex Pluecker coordinate of null momentum factors, its modulus gives the rest energy, its phase acts by internal conjugacy and survives in oriented turn amplitudes, and the

| Construction                | Exact local unitary | Exact finite histories | Mass input                      | Dimension                | Continuum control                           | Machine checked |
|-----------------------------|---------------------|------------------------|---------------------------------|--------------------------|---------------------------------------------|-----------------|
| Feynman checkerboard [4, 5] | not its object      | yes                    | assigned corner weight          | 1 + 1                    | propagator limit                            | no              |
| Dirac QCA [6, 7]            | yes                 | yes                    | automaton parameter             | 1 + 1                    | channel/wave-packet bounds                  | no              |
| Split-step DCA [8]          | yes                 | not central            | tunable coin angle              | 1 + 1                    | Dirac limit                                 | no              |
| Weyl/Dirac QCA [10]         | yes                 | model dependent        | Weyl coupling                   | up to 3 + 1              | relativistic limit                          | no              |
| Tetrahedral Dirac walk [11] | yes                 | not central            | assigned $e^{-im\gamma^0}$ coin | 3 + 1 tetrahedra         | first-order Dirac limit                     | no              |
| Successive-axis walk [18]   | yes                 | not central            | assigned mass parameter         | 3 + 1                    | first-order relativistic limit              | no              |
| This work                   | yes                 | yes                    | $z = \psi \wedge \phi$          | 1 + 1; finite 3 + 1 lift | uniform $O(1/n)$ symbols; three-torus $L^2$ | yes             |

Table 1: Structural comparison, not a priority ranking. The closest QCA works have stronger field-theoretic or higher-dimensional scope. The present distinction is the geometrically generated complex rest operator, the exact kernel dictionary, and their formal verification.

resulting dependency chain is machine checked. Table 1 positions that claim against the closest constructions.

The Pavia programme moreover derives the Weyl and Dirac automata from informational first principles [10]; the derivation offered here is geometric rather than axiomatic, and the two are answers to different questions.

The kinematic layer itself has long roots, and we claim none of it. Weyl–van der Waerden and massive spinor-helicity formalisms routinely decompose a massive momentum into two null spinors, with  $\det K = \lambda_1 \lambda_2 = k^2$  appearing explicitly in the eigen-decomposition of the momentum bispinor [3], and the bold-spinor formalism of Arkani-Hamed, Huang, and Huang [2] makes the two-null-spinor description of massive states the organizing principle of amplitude kinematics. Those formalisms use the decomposition as covariant bookkeeping for scattering amplitudes. The present paper starts where that bookkeeping stops: the wedge coordinate is promoted to the off-diagonal entry of a canonical odd Hermitian operator, the operator is classified (Theorem 3.2), exponentiated into an exactly unitary walk, tracked through directed histories and a local connection, and every finite statement in that chain is kernel-checked. The amplitude formalisms neither assert nor need such a dynamical chain; the overlap is the classical  $2 \times 2$  Cauchy–Binet layer, which we label CLASSICAL throughout.

## Exact consequences for published constructions

Beyond structural comparison, several of this paper’s kernel-checked obstructions bear directly on published proposals. We collect the exact statements here; each is proved in the section cited, and none is a new claim.

- **Stay-put walks (Gupta–Short).** The doubler- and pseudo-doubler-free family of Gupta and Short [16] is exactly of the degree-one Laurent form covered by the stationary-amplitude no-go (53), so its tangent generator cannot be an involution: the stay-put freedom that removes the doublers is purchased exactly by relinquishing the involutory unit-speed Dirac tangent (corollary in the text around Eq. (52); their Eqs. (29)–(30) and (37)). Both programs share the residual open problem their Appendix F records: extraneous Weyl-like low-energy solutions at isolated momenta.
- **Onsite repairs of successive-axis architectures.** For the four-channel, range-one, single-factor-per-axis  $3 + 1$  architecture — the class containing the successive-axis composition formalized here, following [18] — the three even-parity origin aliases are exact, and the all-coins alias theorem (54) proves that *no* momentum-independent onsite coin, the complex Pluecker mass coin included, removes any of them. Extra range, coupled substeps, or enlarged cells are structurally required, not optional. Exact two-step temporal blocking closes only the mass subgroup, so composition-order effects in such architectures are exact, not perturbative artifacts.

- **Scope relative to action-based no-gos.** The Nielsen–Ninomiya theorem [17] governs chirally-invariant local *action/Hamiltonian* discretizations, and the minimally-doubled Karsten–Wilczek/Boriçi–Creutz families live in that class. The obstructions here are the strictly-unitary *discrete-time* analogues (exact one-step unitarity, origin normalization, involutory tangent); neither family of results implies the other, and the two boundaries constrain a  $3 + 1$  regulator from logically independent directions.

## The reparametrization test

At constant nonzero  $z$ , the one-particle spectrum alone cannot distinguish this construction from a Dirac walk whose mass was assigned as  $m = |z|$ . The claim of added content must therefore survive after that spectrum is held fixed. Table 2 states the test explicitly.

| Question                     | Assigned scalar mass                                 | Pluecker-derived input in this paper                                                                                                                                                                                                                       |
|------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constant dispersion          | Same after setting $m =  z $                         | No claimed distinction                                                                                                                                                                                                                                     |
| Gap-closing locus            | External condition $m = 0$                           | Exact collinearity condition $\psi \wedge \phi = 0$                                                                                                                                                                                                        |
| Directed histories           | A real or independently chosen turn coefficient      | Opposite orientations carry $z$ and $\bar{z}$ in the same transfer sum                                                                                                                                                                                     |
| Spatially varying phase      | Additional complex-mass field supplied independently | Primitive-spinor phase induces an exact endpoint connection under local chiral conjugacy                                                                                                                                                                   |
| Finite two-particle probe    | Requires an independent interaction coefficient      | The exhibited quartic transfer reuses the same wedge as amplitudes $z$ and $\bar{z}$ ; equal-modulus fields with conjugate one-particle operators are separated by an exact interference probability ( $4/5$ versus $1$ ) in the supplied pair-kick sector |
| Absolute scale               | Not predicted                                        | Not predicted; the homogeneous-action no-go proves an added scale is required                                                                                                                                                                              |
| Ultraviolet lattice spectrum | Regulator dependent                                  | Still regulator dependent; the present walk retains exact aliases                                                                                                                                                                                          |

Table 2: What is and is not gained by deriving the mass parameter. The quartic row is an explicit finite extension, not yet a theorem that the free action uniquely forces that interaction.

This comparison also fixes the paper’s ceiling. Its centerpiece is a one-particle  $1 + 1$  walk with a new geometric input and a rigorous finite chain; Section 7 adds a controlled finite  $3 + 1$  local extension. It is not yet a new QCA classification, an interacting QCA, or a field theory; its exact high-symmetry audit in fact exposes corner aliases and a mass-independent body-center no-gap obstruction that a successor regulator must remove.

## 2 Null-spinor area and the future cone

### 2.1 From rank-one momenta to area

Let  $\psi_1, \dots, \psi_N \in \mathbb{C}^2$  and define

$$M = \begin{bmatrix} \psi_1 & \cdots & \psi_N \end{bmatrix}, \quad P = MM^\dagger = \sum_{i=1}^N \psi_i \psi_i^\dagger. \quad (1)$$

Each summand is positive semidefinite and has rank at most one. Under the mostly-minus Pauli map, it is a future null four-momentum. The determinant of  $P$  is the Lorentzian square of the total momentum.

**Theorem 2.1** (Null-spinor area identity). *For the matrix in (1),*

$$\det P = \left\| \bigwedge^2 M \right\|_{\text{HS}}^2 = \sum_{i < j} |\det [\psi_i \ \psi_j]|^2. \quad (2)$$

*Consequently  $\det P \geq 0$ , and  $\det P = 0$  if and only if the occupied columns span a subspace of dimension at most one.*

#### CLASSICAL KERNEL

*Proof.* Cauchy–Binet applied to  $MM^\dagger$  expresses its determinant as the sum of the squared moduli of all  $2 \times 2$  minors of  $M$ . Those minors are exactly the coordinates of  $\bigwedge^2 M$ . Every summand is nonnegative, so the sum vanishes exactly when every pairwise wedge vanishes, equivalently when  $\text{rank } M \leq 1$ .  $\square$

Define

$$\mu := \left\| \bigwedge^2 M \right\|_{\text{HS}} = \sqrt{\det P}. \quad (3)$$

The quantity  $\mu$  is the first-order rest energy;  $\mu^2$  is the Lorentzian quadratic invariant. The exterior norm makes the geometric word “area” literal rather than metaphorical.

For readers who prefer spatial directions, write the null momenta as  $p_i = E_i(1, \mathbf{n}_i)$  with  $E_i \geq 0$  and  $|\mathbf{n}_i| = 1$ . With the same mostly-minus normalization,

$$\mu^2 = 2 \sum_{i < j} E_i E_j (1 - \mathbf{n}_i \cdot \mathbf{n}_j). \quad (4)$$

Mass squared is therefore a positive sum of directional disagreements. It is zero only when every occupied direction agrees; no cancellation between different pairs is available.

## 2.2 Why the determinant is canonical

The determinant is not selected merely because it gives the desired answer. On the physical space of Hermitian  $2 \times 2$  matrices it is the unique real quadratic form, up to scale, that vanishes on the full rank-one positive boundary.

**Proposition 2.2** (Null-boundary uniqueness). *Let  $q$  be a real quadratic form on  $\text{Herm}(2)$ . If  $q(\psi\psi^\dagger) = 0$  for every  $\psi \in \mathbb{C}^2$ , then  $q(P) = c \det P$  for some  $c \in \mathbb{R}$ .*

#### CLASSICAL KERNEL

*Proof.* Identify  $P \in \text{Herm}(2)$  with  $(t, \mathbf{x}) \in \mathbb{R}^{1,3}$  so that  $\det P = t^2 - |\mathbf{x}|^2$ . Write  $q(t, \mathbf{x}) = at^2 + 2t \mathbf{b} \cdot \mathbf{x} + \mathbf{x}^T C \mathbf{x}$ , with  $C$  symmetric. On the future null cone, homogeneity reduces the hypothesis to  $q(1, \mathbf{n}) = 0$  for every unit vector  $\mathbf{n}$ . Comparing  $\mathbf{n}$  with  $-\mathbf{n}$  gives  $\mathbf{b} = 0$ . Hence  $\mathbf{n}^T C \mathbf{n} = -a$  on the unit sphere, which forces  $C = -aI$  by polarization. Thus  $q = a(t^2 - |\mathbf{x}|^2) = a \det P$ .  $\square$

The Lean theorem uses the four real coordinates of a Hermitian matrix and nine explicit rank-one spinor probes to eliminate every quadratic coefficient except one multiple of  $pq - r^2 - s^2$ . The older real-symmetric slice and a form that fails the full-boundary hypothesis remain as independent controls.

**Corollary 2.3** (First-order uniqueness). *Suppose  $g$  is nonnegative and homogeneous of degree one on the future cone, and  $g(P)^2$  is a quadratic form vanishing on the rank-one boundary. Then  $g(P) = c\sqrt{\det P}$  for some  $c \geq 0$ .*

CLASSICAL PROOF

This is the logical bridge from a canonical mass-squared invariant to a canonical rest gap. The normalization  $c = 1$  fixes units.

### 3 The Pluecker mass operator

For two spinors, retain the complex orientation data rather than immediately discarding it into an absolute square:

$$z := \psi \wedge \phi = \psi_0 \phi_1 - \psi_1 \phi_0, \quad P := \psi \psi^\dagger + \phi \phi^\dagger. \quad (5)$$

Theorem 2.1 gives  $|z|^2 = \det P$ . Define

$$B_z := \begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}, \quad \Gamma := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (6)$$

**Theorem 3.1** (Pluecker-derived rest operator). *The matrix  $B_z$  is Hermitian and odd for the grading  $\Gamma$ . More precisely,*

$$B_z^\dagger = B_z, \quad \Gamma B_z + B_z \Gamma = 0, \quad B_z^2 = |z|^2 \mathbf{1} = \det(P) \mathbf{1}. \quad (7)$$

For

$$H_z(k) := k\Gamma + B_z, \quad (8)$$

one has

$$\boxed{H_z(k)^2 = (k^2 + \det P) \mathbf{1}}. \quad (9)$$

The rest eigenvalues are  $\pm|z|$ , and  $B_z = 0$  exactly when the two nonzero spinor directions are collinear.

NEW KERNEL

*Proof.* Hermiticity is immediate from the two off-diagonal entries. Direct multiplication gives  $B_z^2 = \text{diag}(z\bar{z}, \bar{z}z) = |z|^2 \mathbf{1}$ , while the diagonal signs in  $\Gamma$  give the anticommutator zero. Expanding  $H_z(k)^2$  leaves  $k^2 \mathbf{1} + B_z^2$  because the mixed term vanishes. Finally  $B_z = 0$  if and only if  $z = 0$ , and a two-spinor wedge vanishes exactly on projective collinearity. Every step of this proof is represented by a general Lean declaration, not by evaluation of a fixed example.  $\square$

Theorem 3.1 is stronger than assigning the scalar  $\det P$  to an otherwise independent Dirac mass slot. The odd operator is constructed from the oriented Pluecker coordinate itself; its square derives the quadratic invariant. The distinction between powers is now fixed:

| Layer                        | Natural quantity                        |
|------------------------------|-----------------------------------------|
| Null-spinor kinematics       | $\mu^2 = \det P =  z ^2$                |
| Odd mass operator            | $B_z$ , with $B_z^2 = \mu^2 \mathbf{1}$ |
| Quadratic costs and Hessians | $\mu^2$                                 |
| Dirac rest energy            | $\mu =  z $                             |
| Finite unitary turn angle    | $a\mu$                                  |
| Exact corner/stay ratio      | $\tan(a\mu)$                            |



### 3.1 Phase covariance and decomposition independence

If  $u \in \mathbb{C}$  has  $|u| = 1$ , let  $W_u = \text{diag}(u, 1)$ . Then

$$W_u B_z W_u^\dagger = B_{uz}, \quad W_u W_u^\dagger = \mathbf{1}. \quad (10)$$

Both identities are kernel-checked. Thus the phase of  $z$  rotates the internal basis but cannot change the rest spectrum.

The construction is also covariant under the spacetime spin group. If  $A \in \text{SL}(2, \mathbb{C})$  acts on both spinors by left multiplication, then

$$(A\psi) \wedge (A\phi) = \det(A)(\psi \wedge \phi) = z, \quad B_{(A\psi) \wedge (A\phi)} = B_z. \quad (11)$$

This is the algebraic  $\text{SL}(2, \mathbb{C})$  statement covering proper-orthochronous Lorentz covariance; it is `KERNEL`. General  $A \in \text{GL}(2, \mathbb{C})$  acts with the expected determinant weight; that weighted extension is an algebraic `PROOF` corollary, not a separate Lean declaration in the present development.

**Theorem 3.2** (Classification of odd Hermitian rest operators). *Let  $B \in M_2(\mathbb{C})$  be Hermitian and satisfy  $\Gamma B + B\Gamma = 0$ . Then there is a unique  $w \in \mathbb{C}$  such that*

$$B = B_w = \begin{pmatrix} 0 & w \\ \overline{w} & 0 \end{pmatrix}, \quad B^2 = |w|^2 \mathbf{1}. \quad (12)$$

Consequently the odd Hermitian square roots of  $\mu^2 \mathbf{1}$  form one phase circle  $w = \mu u$ ,  $|u| = 1$ , and for  $\mu > 0$  they are all related by the conjugacies in (10). Specifying the oriented Pluecker coordinate  $z$  selects one member of that orbit.

NEW `KERNEL` for the normal form and square; `PROOF` for the phase-orbit corollary.

*Proof.* Oddness forces both diagonal entries of  $B$  to vanish, and Hermiticity forces the lower-left entry to be the complex conjugate of the upper-right entry  $w = B_{01}$ . The square is then  $|w|^2 \mathbf{1}$ . If it equals  $\mu^2 \mathbf{1}$ , then  $|w| = \mu$ ; for  $\mu > 0$ , write  $w = \mu u$  with  $|u| = 1$  and apply (10).  $\square$

This also controls the nonuniqueness of the null decomposition. If  $M = [\psi \ \phi]$  has full rank and  $M' = MR$  with  $R \in U(2)$ , then  $M'M'^\dagger = MM^\dagger$  and  $\det M' = (\det R) \det M$ . Since  $|\det R| = 1$ , the two mass operators are related by (10). The unitary-equivalence class of  $B_z$  depends only on  $P$ ; an oriented factorization retains the phase needed by individual turn amplitudes. `CLASSICAL KERNEL`

The distinction matters physically. The spectrum is decomposition independent, while an amplitude can remain sensitive to orientation. The construction preserves exactly the information needed by a path sum and forgets exactly the information a rest energy cannot observe.

### 3.2 One cubic closure: many constituents, and the interaction

The two-spinor operator above is the visible corner of a single finite algebra. For *any* number of spinors  $\psi_i = (u_i, v_i)$ , the pairwise areas  $z_{ij} = u_i v_j - u_j v_i$  assemble into the antisymmetric matrix  $Z = uv^\top - vu^\top$ , and the complete area budget is the Gram determinant, equal by an exact Lagrange identity to the total pairwise area,  $\mu^2 = \sum_{i < j} |z_{ij}|^2$ . The kernel-checked structure (`PlueckerRestOperatorGeneral`) is:

$$ZZ^\dagger Z = \mu^2 Z, \quad B_w = \begin{pmatrix} 0 & Z \\ Z^\dagger & 0 \end{pmatrix}, \quad B_w^3 = \mu^2 B_w, \quad (13)$$

with  $B_w$  odd Hermitian for the block grading,  $B_w^2$  equal to  $\mu^2$  times a projector of trace exactly four for every  $n$  with  $\mu^2 \neq 0$  — the rest gap  $\pm\mu$  always lives on a four-dimensional support — the two-spinor case recovering  $B_z^2 = \mu^2 \mathbf{1}$  on the whole space, and the massless boundary exact:  $Z = 0 \iff \mu^2 = 0$ . The spinor origin of the area data is load-bearing in a precise sense: an explicit non-decomposable antisymmetric control (two disjoint  $2 \times 2$  blocks) satisfies every symmetry hypothesis and still obeys a proportional cubic closure, but with the wrong coefficient —  $ZZ^\dagger Z = 1 \cdot Z$  while its total pairwise area is 2 (witness entry 1 against 2) — so what the spinor origin buys is the *coefficient identification*: the closure constant equals the area budget exactly for spinor-generated data. NEW [KERNEL](#)

The same closure governs the interaction layer. The supplied pair kick of the companion dynamics paper is, up to a global phase, the exact quarter-period pulse of a Hermitian even quartic CAR generator  $K(z)$  obeying

$$K(z)^3 = |z|^2 K(z) \quad (14)$$

on its pair block ([PlueckerPairGenerator](#), [KERNEL](#)), with the gate family closing into an explicit circle group. Equations (13) and (14) have the same finite statement shape — a cubic closure with the squared datum as coefficient — realized once by the odd one-particle rest operator of arbitrarily many null constituents and once by the even many-body coupling that reads the same complex datum. We state this as a coincidence of the shapes of two independently landed theorems, not as a unification: no lemma relates  $B_w$  to  $K(z)$ . The pre-registered selection question — which coupling reads the phase — is resolved (the covariance-forced kick); what remains open is the distinct *dynamical* question, whether the free walk itself forces that interaction, which we list separately in the open problems. What the coincidence does license is a sharper organizing principle for the program: every operator this construction has derived or supplied so far is annihilated by  $X^3 - (\text{area datum}) X$ .

## 4 Exact unitary null histories

### 4.1 Exponentiating the derived rest term

For  $\mu = |z| > 0$ , set  $\hat{B}_z = B_z/\mu$ . Since  $\hat{B}_z^2 = \mathbf{1}$ ,

$$e^{-iaB_z} = \cos(a\mu)\mathbf{1} - i\sin(a\mu)\hat{B}_z. \quad (15)$$

At  $\mu = 0$  the expression is understood by continuity and equals  $\mathbf{1}$ . Combining the turn with the conditional null translation gives the momentum symbol

$$U_a(k; z) := e^{-iak\Gamma} e^{-iaB_z}. \quad (16)$$

Each factor is unitary. The two components translate in opposite null directions; only the odd factor changes direction.

The finite formula in (15) is itself closed under composition: writing it as  $C_z(a)$ , one has

$$C_z(a)C_z(a)^\dagger = \mathbf{1}, \quad C_z(a)C_z(b) = C_z(a+b). \quad (17)$$

The explicit rest eigenvectors, right-unitary decomposition theorem, and this one-parameter group law are all [KERNEL](#).

By phase covariance,  $U_a(k; z)$  is unitarily conjugate to the real-mass walk with parameter  $\mu$ . Its determinant is one and its trace is  $2\cos(ka)\cos(\mu a)$ . Therefore an eigenphase  $e^{-i\omega a}$  obeys

$$\boxed{\cos(\omega a) = \cos(ka)\cos(\mu a)}. \quad (18)$$

At  $k = 0$  the exact eigenphases are  $e^{\mp i a \mu}$ , so the positive principal-branch quasienergy is  $\mu$ . Quasienergy is periodic modulo  $2\pi/a$ ; no global unbranched equality is asserted. The real-mass spectral statement and oriented mass-coin group law are [KERNEL](#); the displayed spectral conjugacy is an immediate corollary of the kernel-checked phase covariance.

## 4.2 The exact corner ratio

Equation (15) gives two kinds of matrix element. A component that keeps its direction contributes  $\cos(a\mu)$ . A direction change contributes  $-i \sin(a\mu)$  times  $z/\mu$  or  $\bar{z}/\mu$ , according to its orientation. Hence an  $n$ -step history  $h$  with  $c(h)$  turns has amplitude

$$A_a(h) = e^{i\Phi_k(h)} \cos(a\mu)^{n-c(h)} (-i \sin(a\mu))^{c(h)} \Xi_z(h), \quad (19)$$

where  $\Phi_k(h)$  is the translation phase and  $\Xi_z(h)$  is a product of unit-modulus orientation factors. Whenever  $\cos(a\mu) \neq 0$ ,

$$A_a(h) = e^{i\Phi_k(h)} \cos(a\mu)^n (-i \tan(a\mu))^{c(h)} \Xi_z(h). \quad (20)$$

**Proposition 4.1** (Unitary checkerboard correspondence). *After removal of the common factor  $\cos(a\mu)^n$ , the exactly unitary walk is a corner-weighted checkerboard sum with finite-spacing corner parameter  $\tan(a\mu)$  and the orientation phase inherited from  $z$ . Under the opposite checkerboard sign convention, the parameter changes sign. In either convention,*

$$\tan(a\mu) = a\mu + O(a^3), \quad (21)$$

but  $\tan(a\mu)$  and  $a\mu$  are not identical at finite spacing.

CLASSICAL for the coin parametrization; NEW [KERNEL](#) for the exact two-weight recursion scaling and the separate transfer-matrix identification; NEW [KERNEL](#) for the directed complex orientation bookkeeping in the same path sum.

*Proof.* Insert the two matrix elements from (15) at every time slice. A history with  $c(h)$  off-diagonal selections and  $n - c(h)$  diagonal selections has exactly the product in (19). Factoring one  $\cos(a\mu)$  from every slice gives (20). The Taylor statement is the standard expansion of the tangent.  $\square$

The Lean development verifies the complete finite dictionary. Its recursive two-weight kernel satisfies

$$K[s, sw]_n = s^n K[1, w]_n,$$

and specialization to  $s = \cos(a\mu)$  and  $w = -i \tan(a\mu)$  gives the unitary checkerboard correspondence exactly when  $\cos(a\mu) \neq 0$ . A quarter-angle nondegenerate witness and the  $a\mu = \pi/2$  pure-flip boundary are included. Separately, the development verifies that finite null histories weighted by a corner power equal a closed binomial kernel and satisfy a discrete Dirac recursion, and that the same direction-resolved history sum equals the corresponding transfer-matrix power. The direct Gaussian-rational path-sum declaration uses a real mass and corner weight  $+i\epsilon m$  in its stated convention. Independently, the directed-edge theorem assigns every history edge the corresponding entry of the actual complex Pluecker mass coin and proves that the complete oriented history sum equals its transfer-matrix power. Thus  $\Xi_z$  in (19) is now packaged directly: opposite turns carry  $z$  and  $\bar{z}$ , with the exact  $3 + 4i$  control proving that the two normalized orientation factors are nonzero and unequal.

The tangent appearance is not asserted as novel by itself. Exponential and Euler-angle parametrizations of unitary Dirac-walk coins were related earlier, including explicit tangent formulas in quantum lattice-Boltzmann work [9]. The contribution here is the exact recursive-kernel scaling law composed with a coin whose generator is derived from the complex Pluecker coordinate, together with its machine verification.

## 5 Full lattice spectrum, symmetry, and doubling audit

Quantum-walk claims cannot be judged only at small momentum. Introduce the dimensionless variables

$$q = ka \in [-\pi, \pi], \quad \theta = \mu a, \quad \Omega = \omega a.$$

The Brillouin zone is periodic in  $q$ , and quasienergy is periodic modulo  $2\pi$ .

**Theorem 5.1** (Complete two-band audit). *The two quasienergy bands are*

$$\Omega_{\pm}(q) = \pm \arccos(\cos q \cos \theta) \pmod{2\pi}. \quad (22)$$

*They are degenerate only when  $q \in \pi\mathbb{Z}$  and  $\theta \in \pi\mathbb{Z}$ . On the physical principal branch  $0 < |\theta| < \pi/2$ , the walk is gapped throughout the Brillouin zone: the positive rest energy at  $q = 0$  is  $|\theta|/a = \mu$ , the interband zero-energy gap is  $2\mu$ , and the circular quasienergy gap at the zone edge is also  $2|\theta|$ . At  $\theta = 0$  there are two massless cones:*

$$(q, \Omega) = (0, 0) \quad \text{and} \quad (q, \Omega) = (\pi, \pi). \quad (23)$$

*For a differentiable band away from a degeneracy,*

$$v_g = \frac{\partial \omega}{\partial k} = \frac{\sin q \cos \theta}{\sin \Omega}, \quad |v_g| \leq 1, \quad (24)$$

*with strict inequality for a massive angle whenever the denominator is nonzero.*

REALIZATION **KERNEL** for the trace-derived dispersion, velocity deficit inequality, exact massless set  $\{0, \pi\}$ , and uniform massive discriminant bound; **PROOF** for the implicit-differentiation identity in (24) and for packaging these facts into the displayed arccos branches.

*Proof.* Equation (22) is the principal solution of (18). A two-band degeneracy requires  $\sin \Omega = 0$ , equivalently  $|\cos q \cos \theta| = 1$ . Each factor has absolute value at most one, so equality forces  $q, \theta \in \pi\mathbb{Z}$ . At  $q = 0$  one has  $\Omega_+ = |\theta|$  on the stated branch; at  $q = \pi$  one has  $\Omega_+ = \pi - |\theta|$ , whose circular distance to the negative band across the quasienergy boundary is  $2|\theta|$ . Implicit differentiation gives (24). Finally,

$$\sin^2 \Omega - \sin^2 q \cos^2 \theta = 1 - \cos^2 \theta = \sin^2 \theta \geq 0,$$

which proves the speed bound and its strict massive form.  $\square$

**Doubling and pseudo-doubling verdict.** The high-momentum  $(\pi, \pi)$  cone is a Floquet zone-edge partner of the low-energy cone. It is not a second zero-quasienergy root, but it is a second relativistic band-touching point in the full lattice spectrum. Following the terminology of Gupta and Short [16], it is naturally classified as a pseudo-doubler: a high-quasienergy mode with a Dirac-like local dispersion. The elementary walk therefore does *not* establish a no-doubling or no-pseudo-doubling theorem. The bounded continuum estimate in Section 6 isolates fixed physical momenta as  $a \rightarrow 0$ , sending  $q \simeq \pi$  to cutoff momentum; that decoupling is not the same as removing the finite-lattice partner.

$$\text{Exact bands: } \cos \Omega = \cos q \cos \theta$$

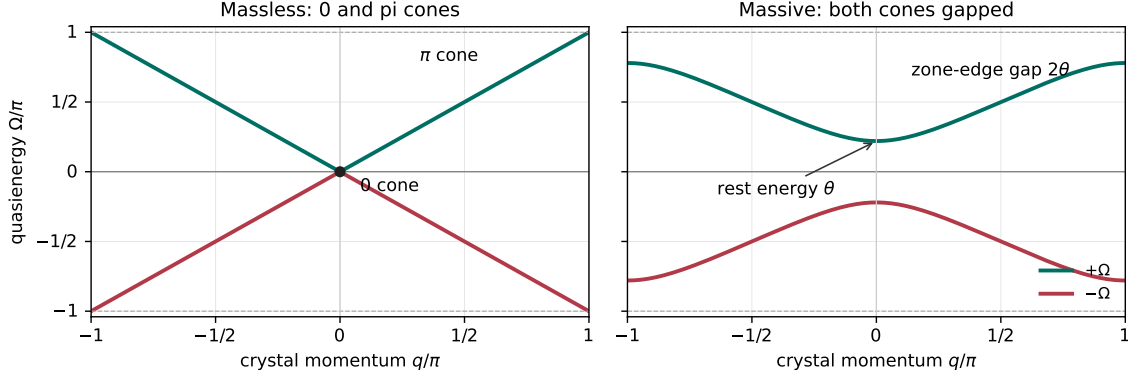


Figure 2: Exact quasienergy bands from  $\cos \Omega = \cos q \cos \theta$ . The left panel shows the massless cone at the origin and its Floquet partner at quasienergy  $\pi$ . A nonzero Pluecker area opens both gaps (right). The curves evaluate the exact formula; they are not a fit or numerical simulation. Generated by `Scripts/sim/plot_null_spinor_dirac_bands.py`.

**Chiral time frame.** Let

$$U_s(q, \theta) = e^{-i\theta\sigma_x/2} e^{-iq\sigma_z} e^{-i\theta\sigma_x/2}.$$

It is unitarily equivalent to  $U(q, \theta)$  and obeys the exact chiral relation

$$\sigma_y U_s(q, \theta) \sigma_y = U_s(q, \theta)^\dagger. \quad (25)$$

Thus the reciprocal pairing of quasienergies is structural, not an artifact of the trace calculation. Equation (25) is the symmetry statement used here; a finer antiunitary or Floquet topological classification is left to a separate analysis [12].

**Robustness.** If  $z$  is perturbed to  $z + \delta z$ , then

$$||z + \delta z| - |z|| \leq |\delta z|.$$

Therefore every perturbation with  $|\delta z| < |z|$  remains off the collinear gap-closing locus and retains a rest energy at least  $|z| - |\delta z|$ . The phase may rotate the internal basis, but by (10) it cannot change a constant-coefficient spectrum.

## 6 Uniform many-step Dirac control

The finite walk should approximate the continuous Dirac flow over a fixed macroscopic time, not merely at one infinitesimal step. For the real representative of the phase-conjugacy class, write

$$U(q, r) = e^{-iq\sigma_z} e^{-ir\sigma_x}, \quad H(k, \mu) = k\sigma_z + \mu\sigma_x, \quad V_t(k, \mu) = e^{-itH(k, \mu)}. \quad (26)$$

**Theorem 6.1** (Uniform bounded-momentum product limit). *For  $K, M \geq 0$ , define*

$$C_{\text{box}}(K, M) = 2K^2 + 2M^2 + KM^2 + K^2M + KM, \quad (27)$$

$$D_{\text{box}}(K, M) = 4C_{\text{box}}(K, M) + 4(K + M)^2 e^{K+M}. \quad (28)$$

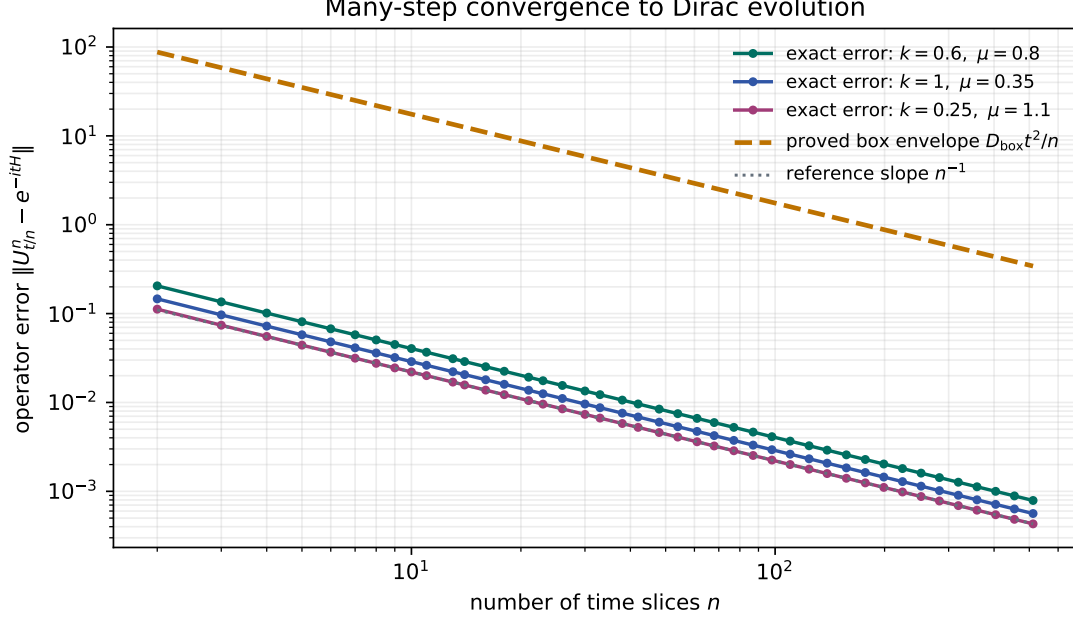


Figure 3: Exact many-step operator errors for three nondegenerate parameter choices. All follow the proved  $n^{-1}$  rate and remain below the explicit compact-box envelope. The points evaluate the exact finite matrices; the envelope and convergence statement are kernel checked.

If  $|k| \leq K$ ,  $|\mu| \leq M$ ,  $n > 0$ , and  $|t/n| \leq 1$ , then

$$\left\| U\left(k\frac{t}{n}, \mu\frac{t}{n}\right)^n - V_t(k, \mu) \right\| \leq D_{\text{box}}(K, M) \frac{t^2}{n}. \quad (29)$$

The bound is uniform on the displayed parameter box, and its right-hand side tends to zero as  $n \rightarrow \infty$ .

#### NEW **KERNEL**

*Proof architecture.* The one-step walk differs from  $\mathbf{1} - i(t/n)H$  by an explicit quadratic remainder. The exact exponential differs from the same first-order matrix by another quadratic remainder, giving  $\|U - V_{t/n}\| \leq D(k, \mu)t^2/n^2$ . Both matrices are unitary, so the power telescope has no norm-growth factor:

$$\left\| U^n - V_{t/n}^n \right\| \leq n \|U - V_{t/n}\|.$$

The exact short-time flows compose to  $V_t$ , producing the  $1/n$  rate. Finally  $D(k, \mu) \leq D_{\text{box}}(K, M)$  on the parameter box. Lean checks the entrywise remainder estimates, operator-norm conversion, unitarity, telescope, flow composition, box inequality, and limit.  $\square$

This proof is a finite-dimensional first-order product-formula argument. Its constant is explicit rather than optimized; sharper general Trotter-error theory provides context for that distinction [19].

For  $z = \mu u$  with  $|u| = 1$ , the diagonal unitary  $W_u$  commutes with  $\Gamma$  and conjugates  $B_\mu$  to  $B_z$ . Unitary invariance of the operator norm therefore transfers (29) to the oriented Pluecker walk. This one-line conjugacy corollary is **PROOF** pending a combined declaration.

The theorem proves uniform convergence of the finite momentum symbol in  $1 + 1$  dimensions. It is not yet an infinite-volume inverse-Fourier theorem, a position-space PDE convergence theorem, or an interacting limit. Those quantifiers are separate problems, not implications hidden in the word “continuum.”

## 7 Finite position kernels and a controlled $3 + 1$ extension

The uniform estimates already imply a finite position-space statement. Let  $\mathcal{K}$  be a finite momentum set, let  $\chi_x(k)$  be phases with  $|\chi_x(k)| \leq 1$ , and define the synthesized matrix kernel

$$\mathcal{S}_x[A] := \sum_{k \in \mathcal{K}} \chi_x(k) A(k).$$

Applying the triangle inequality to Theorem 6.1 gives the kernel-checked bound

$$\|\mathcal{S}_x[U_n] - \mathcal{S}_x[V_t]\| \leq |\mathcal{K}| D_{\text{box}}(K, M) \frac{t^2}{n}. \quad (30)$$

This controls every finite periodic Fourier kernel. The cardinality factor is the generic price of applying the triangle inequality pointwise. It disappears at the level of normalized  $L^2$  wave packets. For Mathlib’s unnormalized vector-valued transform  $\mathcal{F}$  on  $\mathbb{Z}/N\mathbb{Z}$ , we prove

$$\sum_k \|(\mathcal{F}f)(k)\|^2 = N \sum_x \|f(x)\|^2, \quad \sum_x \|\mathcal{F}^{-1}g(x)\|^2 = \frac{1}{N} \sum_k \|g(k)\|^2. \quad (31)$$

Consequently, if a modewise error obeys  $\|\delta(k)\| \leq \varepsilon \|c(k)\|$ , then

$$\sum_x \|\mathcal{F}^{-1}\delta(x)\|^2 \leq \frac{\varepsilon^2}{N} \sum_k \|c(k)\|^2. \quad (32)$$

Both identities and the implication are `CLASSICAL KERNEL`; a one-mode packet is included as an exact normalization control. We also prove the operator-valued corollary: a uniform bound  $\|A(k) - B(k)\| \leq \varepsilon$  implies (32) for  $\delta(k) = (A(k) - B(k))c(k)$ . Applying that corollary to the local walk uses the exact plane-wave block stated below. Iterating the transform in the three coordinates gives, for  $N \geq 1$ , the kernel-checked product-torus identity

$$\sum_{x \in (\mathbb{Z}/N\mathbb{Z})^3} \|\mathcal{F}_3^{-1}g(x)\|^2 = \frac{1}{N^3} \sum_{k \in (\mathbb{Z}/N\mathbb{Z})^3} \|g(k)\|^2. \quad (33)$$

Hence, again for  $N \geq 1$ , a uniform modewise operator error  $\varepsilon$  gives the exact wave-packet coefficient  $\varepsilon^2/N^3$ , with no cardinality loss. The momentum-sign bridge is also kernel-checked: for the positive-exponential convention the finite character block equals the analytic block at  $k_j\varepsilon = -2\pi n_j/L$ , including an exact quarter-zone  $-i$  sign control.

There is also a countable successor to the finite Plancherel statement. If  $e_n(k)$  is a countable family of mode errors such that  $e_n(k) \rightarrow 0$  for every  $k$  and  $\|e_n(k)\|^2 \leq b(k)$  for one summable envelope  $b$ , then

$$\sum_k \|e_n(k)\|^2 \longrightarrow 0. \quad (34)$$

This is a direct vector-valued Tannery specialization, proved in Lean with a nonzero one-mode control. It isolates the remaining physical obligation: construct the walk-specific square-summable



envelope and identify the limiting multiplier with the Dirac PDE propagator under a continuum Fourier isometry.

The corresponding measure-theoretic multiplier step is also closed. If  $f \in L^2$  and pointwise errors satisfy  $\|e_n(k)\| \leq \varepsilon_n \|f(k)\|$  almost everywhere, where  $0 \leq \varepsilon_n$  and  $\varepsilon_n \rightarrow 0$ , then

$$\|e_n\|_{L^2} \rightarrow 0. \quad (35)$$

The proof is a kernel-checked ‘eLpNorm’ monotonicity-and-squeeze argument. The walk-specific compact-support application is now closed as well. For any fixed  $t \in \mathbb{R}$ , suppose  $f \in L^2$  and, almost everywhere on the support of  $f$ ,  $|k_x|, |k_y|, |k_z| \leq K$  and  $|z| \leq M$ , with  $K, M \geq 0$ . Applying the complex split symbol for  $n + 1$  steps gives an error multiplier  $E_n$  satisfying

$$\|E_n f\|_{L^2} \rightarrow 0, \quad \|E_n(k)f(k)\| \leq \frac{D_{K,M} t^2}{n+1} \|f(k)\|. \quad (36)$$

The displayed pointwise envelope holds for all sufficiently large  $n$ ; the finite prefix is irrelevant to the sequence limit. This walk-specific composition is NEW **KERNEL**. What remains is the lattice/continuum Fourier-isometry and identification of the limiting multiplier with the position-space Dirac PDE, not compact-support multiplier convergence.

We now also close the changing-scale multiplier gate in  $3 + 1$  dimensions. At refinement level  $N$ , set

$$h_N = \frac{1}{N+1}, \quad K_N = (N+1)^2, \quad W_N = 3(N+1) + M + 1, \quad qquad n_N = W_N^4. \quad (37)$$

The momentum grid spacing tends to zero, while its represented center radius is exactly  $h_N K_N = N + 1$  and therefore exhausts momentum space. For every integer mode  $k_N$  in the complete cutoff box and every fixed mass  $|m| \leq M$ , the live four-component split step obeys the uniform bound

$$\left\| U_{h_N k_N} \left( \frac{t}{n_N} \right)^{n_N} - e^{-itH(h_N k_N, m)} \right\| \leq \frac{2t^2}{W_N^2} \exp \left( \frac{|t|}{W_N^3} \right) \rightarrow 0. \quad (38)$$

This changing-spacing, exhausting-box theorem is NEW **KERNEL**. Its boundary fixture takes the positive  $x$ -face mode, whose represented physical momentum is exactly  $N + 1$ ; the result is therefore not a disguised fixed-compact-momentum limit.

The concrete cell-sampling dense core is now closed as well. Let  $f$  be a complex Lipschitz momentum field with constant  $L$ , and let a finite family of the half-open cells of side  $h_N$  cover its support. Sampling  $f$  at each cell center gives the exact global estimate

$$\int_{\mathbb{R}^3} |I_N f(k) - f(k)|^2 dk \leq \#S_N h_N^3 \left( \frac{L h_N}{2} \right)^2. \quad (39)$$

Consequently, if the represented cell volume  $\#S_N h_N^3$  is uniformly bounded and  $h_N \rightarrow 0$ , the left side tends to zero. Cell radius  $h_N/2$  in the product sup norm, exact union volume, support reduction, and the limit are all NEW **KERNEL**. This is strong  $L^2(\mathbb{R}^3)$  convergence for a compact-support Lipschitz dense core. The regularity bridge to arbitrary data is also now kernel-checked: every complex  $L^2(\mathbb{R}^3)$  field admits a compactly supported smooth approximant with arbitrarily small  $L^2$  error, and that same approximant has one global Lipschitz constant. Compact support is load-bearing: the smooth noncompact quadratic  $x \mapsto x_0^2$  has no global Lipschitz constant.

There is a sharp representative-level obstruction to extending this center sampler to arbitrary  $L^2$ . A function equal to one at a single cell center and zero elsewhere is zero almost everywhere, but center sampling reads the value one; sampling the AE-equal zero function reads zero. Thus



the center sampler is not AE-invariant and cannot define a bounded operation on  $L^2$  equivalence classes. This no-go is NewResult Kernel, not merely an analytic warning. Normalized cell averaging supplies the correct first repair: it is AE-invariant, sends the one-point spike to zero, and returns exactly one on the constant-one function for every positive cell size.

The representative-safe repair has now advanced substantially. The finite cell-average projection is an exact  $L^2$  contraction; compact support is eventually covered by the explicit expanding schedule; only the active cells, not the full growing box, have a uniform volume bound; and the exact three-term transfer estimate has constants 6 and 3. The final epsilon composition to arbitrary  $L^2$  data is a separate capstone now under kernel verification. Beyond it remain the identification of cell averages with coefficients evolved by the live walk, composition with (38), and inverse continuum Fourier identification of the position-space Dirac flow. We do not yet claim that final PDE theorem.

The same architecture extends to four internal components. The successive-axis factorization itself is the standard construction used by Mlodinow and Brun [18]; what is new here is the Pluecker-derived complex coin, the exact finite-torus implementation with a two-sided inverse and strict causal cone, and the kernel-checked all-zone audit that follows. Let Hermitian matrices  $\alpha_1, \alpha_2, \alpha_3, \beta \in M_4(\mathbb{C})$  satisfy

$$\alpha_i^2 = \beta^2 = \mathbf{1}, \quad \{\alpha_i, \alpha_j\} = 0 \ (i \neq j), \quad \{\alpha_i, \beta\} = 0,$$

and set

$$H_4(\mathbf{k}, m) = \sum_{j=1}^3 k_j \alpha_j + m \beta.$$

The explicit matrices used in the artifact obey

$$H_4(\mathbf{k}, m)^2 = (|\mathbf{k}|^2 + m^2) \mathbf{1}. \quad (40)$$

For a substep  $\varepsilon$ , define

$$U_\varepsilon^{(4)}(\mathbf{k}, m) = e^{-i\varepsilon k_x \alpha_1} e^{-i\varepsilon k_y \alpha_2} e^{-i\varepsilon k_z \alpha_3} e^{-i\varepsilon m \beta}. \quad (41)$$

The order in (41) is not cosmetic: the spatial generators do not commute, and the artifact contains that noncommutation as a negative control against a false second-order claim.

**Theorem 7.1** (Finite local 3 + 1 walk and compact rate). *On a finite periodic three-dimensional position register, there is an explicit successive-axis update built from one-site conditional shifts and the mass coin in (41). It preserves the finite inner product exactly, has an explicit two-sided inverse, and reads only sites reached by one conditional source shift along each axis in one cycle. Its product plane-wave sectors are invariant, and on each sector the update is exactly the ordered axis-0/axis-1/axis-2/mass matrix block built from finite roots of unity. For the positive-exponential plane-wave convention, that block is (41) when*

$$k_j \varepsilon = -\frac{2\pi n_j}{L}. \quad (42)$$

If  $|k_x|, |k_y|, |k_z| \leq K$ ,  $|m| \leq M$ ,  $n > 0$ , and  $|t/n| \leq 1$ , then

$$\left\| \left( U_{t/n}^{(4)}(\mathbf{k}, m) \right)^n - e^{-itH_4(\mathbf{k}, m)} \right\| \leq 16(3K + M)^2 e^{3K+M} \frac{t^2}{n}. \quad (43)$$

Every finite Fourier synthesis obeys the same bound multiplied by its number of modes, while the normalized product-torus  $L^2$  wave-packet error obeys the squared bound divided by  $N^3$ . Moreover, let

$B_4(z)$  be the phase-retaining rest operator and let  $U_\varepsilon^{(4)}(\mathbf{k}, z)$  be the chiral conjugate of the displayed step at  $m = |z|$ . Its exact comparison flow is generated by

$$H_4(\mathbf{k}, z) = \sum_{j=1}^3 k_j \alpha_j + B_4(z), \quad H_4(\mathbf{k}, z)^2 = (|\mathbf{k}|^2 + |z|^2) \mathbf{1},$$

and whenever  $|z| \leq M$  it satisfies the same bound

$$\left\| \left( U_{t/n}^{(4)}(\mathbf{k}, z) \right)^n - e^{-itH_4(\mathbf{k}, z)} \right\| \leq 16(3K + M)^2 e^{3K+M} \frac{t^2}{n}. \quad (44)$$

REALIZATION **KERNEL** for exact local norm preservation, Clifford square, finite-character plane-wave block, compact rate, finite synthesis bound, and the trigonometric/sign conversion (42) to the analytic symbol; NEW **KERNEL** for the complex-phase transfer (44) and the iterated three-torus normalization (33).

*Proof architecture.* Each  $\alpha_j$  has eigenvalues  $\pm 1$ . An explicit unitary eigenbasis turns the corresponding diagonal one-site conditional shift into the physical Clifford factor; composing those shifts with the mass coin proves exact norm preservation. Direct substitution into every product plane wave extracts one unit-modulus character per conditional shift and proves the exact ordered finite block without an asymptotic argument. The closed form  $e^{-iq\alpha_j} = \cos q \mathbf{1} - i \sin q \alpha_j$  identifies the ordered analytic matrix symbol after the sign convention in (42). A four-factor quadratic remainder estimate, exact unitarity, and the unitary power telescope then give (43). Applying the finite synthesis theorem gives its position-kernel corollary.  $\square$

For  $z \neq 0$ , the phase transfer is exact. If  $W_z$  is the chiral unitary at the polar angle of  $z$ , then

$$W_z H_4(\mathbf{k}, |z|) W_z^\dagger = H_4(\mathbf{k}, z), \quad W_z e^{-itH_4(\mathbf{k}, |z|)} W_z^\dagger = e^{-itH_4(\mathbf{k}, z)}.$$

The operator norm cannot increase under this conjugation, so the real-mass estimate transfers without changing its constant.

### The 3 + 1 high-symmetry verdict

The continuum tangent is exactly isotropic:  $H_4(\mathbf{k}, 0)^2 = |\mathbf{k}|^2 \mathbf{1}$ . The finite cubic walk is not globally doubler-free. At the eight corners  $(q_x, q_y, q_z) \in \{0, \pi\}^3$ , its massless symbol is

$$U^{(4)}(q_x, q_y, q_z, 0) = (-1)^r \mathbf{1}, \quad r = \#\{j : q_j = \pi\}. \quad (45)$$

Thus  $(\pi, \pi, 0)$ ,  $(\pi, 0, \pi)$ , and  $(0, \pi, \pi)$  are distinct momenta with exactly the same zero-quasienergy symbol as the origin. Equation (45), all three distinctness witnesses, and radial isotropy of the continuum square are REALIZATION **KERNEL**. This is an actual finite-regulator obstruction, not a caveat inferred from a plot.

The massive walk has a stronger interior obstruction. At body-center momentum  $\mathbf{q}_B = (\pi/2, \pi/2, \pi/2)$ , let  $u_\theta = \cos \theta + i \sin \theta$  and

$$v_+(\theta) = (-u_\theta, 0, 1, 0)^T, \quad v_-(\theta) = (u_\theta, 0, 1, 0)^T.$$

For every real mass angle  $\theta$ , the exact finite step obeys

$$U^{(4)}(\mathbf{q}_B, \theta) v_+(\theta) = v_+(\theta), \quad U^{(4)}(\mathbf{q}_B, \theta) v_-(\theta) = -v_-(\theta), \quad v_\pm(\theta) \neq 0. \quad (46)$$

Equivalently, the body-center step is a non-scalar involution for all  $\theta$ . It therefore retains exact quasienergies 0 and  $\pi$  even after the mass coin is turned on: the present cubic regulator admits no uniform Floquet gap (a body-center crossing at 0 and  $\pi$  persists for every mass angle). The closed body-center matrix, involution identity, and both explicit eigenmodes are NO-GO [KERNEL](#).

The interior crossings moreover carry an exact charge bookkeeping. At each of the eight nodes  $q_j \in \{\pi/2, 3\pi/2\}$  the massive step has two-dimensional  $+1$  and  $-1$  eigenspaces simultaneously, and the Schur-reduced crossing Jacobian of every node and gap is one of exactly two diagonal matrices,  $\pm \frac{4}{5} \text{diag}(-1, 1, -1)$ , with determinant  $\pm(\cos \theta)^3$  at the 3-4-5 angle — every node is a nondegenerate cone. The resulting local charges (the Jacobian determinant signs) are distributed by the parity of the node's  $3\pi/2$ -coordinates, are opposite at the two quasienergies of each node (the Floquet pairing), and sum to zero at each quasienergy separately. This is now machine-checked from the walk symbol itself. The base census arithmetic — the two Jacobian determinants, the parity-indexed assignment, and its sum-zero — is kernel-checked in `SplitStepChargeBalance`; and the derivation of the Jacobians *from the Bloch symbol* is kernel-checked for all eight nodes in `SplitStepSchurJetAllNodes`: the exact node collapse  $U_n = (-1)^{\text{par}(n)} U_0$  (from  $\cos \frac{3\pi}{2} = 0$ ,  $\sin \frac{3\pi}{2} = -1$ ), the two-dimensional  $\pm 1$  kernels at every node, the Schur-reduced  $3 \times 3$  Jacobian, and the resulting per-node Floquet opposition and sum-zero. The derivation is anchored to the landed census by a compiler-enforced bridge (`CensusDerivationBridge`): the walk-derived assignment is proved equal, at every node and gap, to the `SplitStepChargeBalance` census imported from that module, so nothing rests on an unchecked copy. The one modelling choice made explicit is that the nondegenerate-cone reading uses the normal-matrix structure of each node. The doublers of this walk are not merely present: they are exactly the charge partners the sum rule demands — derived from the symbol — and the two gaps compensate node by node.

The obstruction is also visible in an exact all-zone criterion. Write  $x = \cos q_x$ ,  $y = \cos q_y$ ,  $w = \cos q_z$ , and  $c = \cos \theta$ , and define

$$S = 4c^2x^2y^2w^2 - 2c^2x^2y^2 - 2c^2x^2w^2 + c^2x^2 - 2c^2y^2w^2 + c^2y^2 + c^2w^2 - 2x^2y^2w^2 + x^2y^2 + x^2w^2 + y^2w^2, \quad (47)$$

$$P_0 = S - 2cxyw, \quad P_\pi = S + 2cxyw. \quad (48)$$

For the live ordered successive-axis symbol, the finite determinant expansion is now exact at every real momentum and mass angle:

$$\det(U^{(4)} - \mathbf{1}) = 4P_0, \quad \det(U^{(4)} + \mathbf{1}) = 4P_\pi. \quad (49)$$

Thus  $P_0 = 0$  and  $P_\pi = 0$  are respectively necessary and sufficient for zero- and  $\pi$ -quasienergy modes. At body center  $x = y = w = 0$ , both polynomials vanish for every  $\theta$ , recovering (46). The two determinant identities, their explicit  $4 \times 4$  expansion, and the definitional bridge to the walk used in the continuum estimate are NO-GO [KERNEL](#).

The zero sets are now classified on the principal massive branch. If  $0 < |c| < 1$  and  $|x|, |y|, |w| \leq 1$ , then

$$P_0 = 0 \iff x = y = w = 0 \iff P_\pi = 0. \quad (50)$$

The proof uses the exact decomposition

$$P_0 = (cx - yw)^2 + (y^2 + w^2 - 2y^2w^2)(c^2 + x^2(1 - 2c^2)), \quad (51)$$

whose second factor is strictly positive and whose first prefactor is nonnegative on this branch. Equality then forces all three momentum cosines to vanish; the  $P_\pi$  statement follows by  $c \mapsto -c$ .

Thus the simultaneous body-center cosine locus is the complete zero- and  $\pi$ -quasienergy crossing set for every principal massive angle, not merely a sampled obstruction. The hypotheses are load-bearing: at  $c = 0$ , additional zeros occur, including  $(x, y, w) = (1, 0, 0)$ . The classification and this boundary control are NO-GO [KERNEL](#). They classify the failure of the present regulator; they do not make it alias-free.

**A degree-one stationary-amplitude no-go.** The most economical proposed repair is to add a stay-put channel to each nearest-neighbor factor. The exact ansatz is

$$F(q) = e^{iq}A + B + e^{-iq}C. \quad (52)$$

Let  $M$  be any Hermitian involution. If  $F(q)$  is unitary for every real  $q$ ,  $F(0) = \mathbf{1}$ , and  $F'(0) = -iM$ , then

$$B = 0. \quad (53)$$

The proof evaluates the exact unitarity residual at  $q = \pm\pi/2$ ; adding the two resulting matrix identities forces  $B^\dagger B = 0$ . Specializing  $M = \alpha_j$  rules out a nonzero stationary amplitude in a degree-one factor with the full Dirac tangent, even before zone-edge separation is imposed. All three specializations to the live  $3 + 1$  generators are separately kernel-checked. This is NO-GO [KERNEL](#). A nonvacuity control shows that a partial-channel Hermitian tangent with a stationary kernel does admit an exactly unitary degree-one witness with  $B \neq 0$  and a non-scalar zone edge. That witness does not retain the full Dirac tangent. Thus stationary amplitudes remain viable only after a genuine change of architecture, such as a larger internal cell, longer Laurent range, multistep factorization, or a non-involutory tangent.

The last route is not hypothetical. Gupta and Short [16] construct doubler- and pseudo-doubler-free nearest-neighbor one-step unitary walks, in  $1 + 1$  and  $3 + 1$  dimensions, whose enabling resource is exactly a nonzero stay-put amplitude  $\gamma_0 = \mathbf{1} - \gamma_+ - \gamma_-$  built from tilted, non-orthogonal velocity projectors (their Eqs. (29)–(30) and (37)). Their step is exactly of the degree-one Laurent form (52), unitary at every momentum and normalized at the origin, with stationary block  $\gamma_0$ . The no-go (53) therefore yields a corollary about their family: its tangent generator, though automatically Hermitian for a unitary origin-normalized step, cannot be an involution ( $M^2 \neq \mathbf{1}$ ). The stay-put freedom that removes the doublers is purchased exactly by relinquishing the involutory unit-speed Dirac tangent, and our partial-channel escape witness realizes the same mechanism in kernel-checked form. Their  $3 + 1$  family moreover retains extraneous Weyl-like low-energy solutions at isolated momenta  $\pm \mathbf{q}(\theta)$ , which the authors suggest a more general higher-dimensional construction might remove (their Appendix F), so a strictly local, exactly unitary, fully alias-free  $3 + 1$  walk with the complete involutory Dirac tangent remains open in both programs.

The obstruction is stronger for a separable three-axis product. Once  $B_x = B_y = B_z = 0$ , origin normalization forces each factor to obey  $F_j(\pi) = -\mathbf{1}$ . Therefore, for any momentum-independent onsite coin  $Q$ , including the complex Pluecker mass coin,

$$U_Q(0, 0, 0) = U_Q(\pi, \pi, 0) = U_Q(\pi, 0, \pi) = U_Q(0, \pi, \pi). \quad (54)$$

This all-coins alias theorem is NO-GO [KERNEL](#). It proves that changing the onsite mass data cannot repair the cubic partners inside the four-channel, range-one, single-factor-per-axis architecture. Extra range, coupled substeps, or enlarged/ancillary cells are not optional embellishments; at least one is structurally required.

An attempted commutator–Wilson repair sharpens that requirement. Consider the exactly unitary one-axis gate

$$W_r(q) = F_\beta(rq)F_{\gamma_5}(rq)F_\beta(-rq)F_{\gamma_5}(-rq), \quad F_A(s) = \cos s \mathbf{1} - i \sin s A. \quad (55)$$

Its linear term cancels, so at the Bloch-matrix level it appears to supply a second-order Wilson-like correction without disturbing the Dirac tangent. However, in the fixed one-cell ansatz strict finite Laurent range requires the frequency  $r$  to be an integer. For every integer  $r$  the exact zone-edge identity is

$$W_r(\pi) = \mathbf{1}, \quad (56)$$

and all three even-corner aliases in (54) remain exactly equal to the origin. Conversely, the noninteger trigonometric values that lift selected corners are not  $2\pi$ -periodic finite Laurent polynomials in  $e^{iq}$  and therefore do not define the claimed strict local one-cell QCA. This family-scoped locality/dealiasing incompatibility, including the range-one control, is NO-GO [KERNEL](#). It does not exclude a larger cell or a genuinely different finite product; it prevents a nonlocal Bloch control from being mislabeled as that successor.

The matching cross-architecture algebraic invariant is now exact in one Laurent variable. The determinant of every invertible finite Laurent matrix is a nonzero scalar times a uniquely exponentiated monomial, and that exponent is additive under composition, vanishes on the identity, and equals  $n$  on the one-channel shift  $T^n$ . This is the algebraic precursor of the translation-invariant one-particle quantum-walk index in GNVW Proposition 5 [13]; identifying the two still requires a complex-unitary Laurent/Fourier representation theorem and an explicit shift-sign convention. It is not the positive-rational many-body GNVW QCA index, an entropy-flow theorem, or a no-doubling result.

There is also a necessary correction to the charge strategy. A complete mass-admitting four-component Dirac crossing admits an explicit gapped algebraic contraction to a constant. In the standard class-A topological reading this makes the complete crossing neutral, but that topological bridge is not itself part of the kernel result. If  $F_0(\mathbf{n}) = \sum_j n_j \alpha_j$  is the flattened tangent and  $\beta$  anticommutes with it, then

$$F_s(\mathbf{n}) = \cos(\pi s/2) F_0(\mathbf{n}) + \sin(\pi s/2) \beta \quad (57)$$

satisfies  $F_s(\mathbf{n})^2 = \mathbf{1}$  and ends at the constant matrix  $\beta$ . The finite matrix identity, including explicit anticommuting involutions and a missing-anticommutation control, is [REALIZATION KERNEL](#). The nonzero local charges instead belong to the two Weyl sectors selected by  $\Xi = -i\alpha_1\alpha_2\alpha_3$ . For the actual live cubic Clifford tangent, an exact finite calculation constructs two orthogonal Weyl bases and restricts  $(\alpha_1, \alpha_2, \alpha_3)$  to the Pauli triple and its negative. The resulting supplied-Jacobian charges are exactly  $+1$  and  $-1$ , so their four-component sum vanishes; a singular tangent has charge zero. This is [REALIZATION KERNEL](#) at the local tangent. A sectorwise Brillouin-zone argument still requires the additional global hypothesis  $[U(\mathbf{q}), \Xi] = 0$  at every momentum; local chirality at the origin is insufficient. For the actual ordered Bloch step this gate is now exact:

$$[U(\mathbf{q}, \theta), \Xi] = 0 \quad \Longleftrightarrow \quad \sin \theta = 0. \quad (58)$$

The chirality is a Hermitian involution, and its positive-sector projector commutes with the complete massless walk at every momentum. The walk is therefore globally Weyl-sector split, while every genuine mass angle ( $\sin \theta \neq 0$ ) mixes the sectors; an exact quarter-turn fixture gives a nonzero commutator. Here the mass weight is  $\sin \theta$ ; a cosine variable used in the separate determinant factorization is not that weight. These statements are [NEW KERNEL](#). They determine precisely when the sectorwise no-go can be invoked, but it does not supply the missing global charge-sum theorem.

The analytic and arithmetic core of that census is now machine-checked on the  $2 \times 2$  positive-Weyl block. Lean extracts the exact Pauli vector of the live ordered step, proves its complete real Fréchet derivative, and identifies the derivative matrix with the one used in the finite census. Its determinant

obeys

$$\det J(\mathbf{q}) = u_0(\mathbf{q})(\cos^2 q_y - \sin^2 q_y), \quad (59)$$

evaluates to  $\pm 1$  at the sixteen designated principal-torus nodes, and the actual live-Jacobian charges cancel separately on the displayed  $U = +1$  and  $U = -1$  lists. Every designated node consequently has a distinct nondegenerate partner inside its finite list. These are NEW **KERNEL**. A second exact theorem classifies the complete massless  $4 \times 4$  zero- and  $\pi$ -determinant sets in cosine coordinates: each is the appropriate parity class of cube corners together with the body-center locus. The branch-sensitive bridge is now exact. For the actual positive  $2 \times 2$  Weyl block,  $U = +1$  holds if and only if either all three cosine squares are one with cosine product  $+1$ , or all three cosines vanish with sine product  $-1$ . Likewise,  $U = -1$  holds exactly on the odd-parity cosine corners or the body centers with sine product  $+1$ . These two global trigonometric loci are disjoint, and the actual Fréchet-Jacobian determinant is nonzero at every point of either locus. The rank-deficient quarter-axis control lies on neither branch. This is NewResult Kernel and upgrades the displayed charge census from a selected list to an exhaustive positive-Weyl branch classification. It is still a Jacobian-sign doubling theorem, not a formal Chern-number computation. At a full  $4 \times 4$  body center both quasienergies remain present across the two Weyl blocks, which is why the branch theorem could not be inferred from the full determinant set alone.

This distinction sharpens both the no-go and the escape. Read's algebraic  $K$ -theory calculation for complex Laurent automorphisms retains only the constant unit and coordinate-delay classes, with no higher-dimensional strong component in the change-of-rings image [14]. Together with Floquet charge-bulk bookkeeping [15], this supplies a source-supported route to a global-chirality doubling theorem, but the finite-rank and zero/ $\pi$  convention composition is not yet kernel-checked here. Conversely, the exact decomposition  $U_\perp = (U - \Xi U \Xi)/2$  proves that  $U_\perp$  has zero constant and linear coefficients when the normalized Dirac first jet commutes with  $\Xi$ . Therefore the first construction class worth testing consists of exact Laurent-unitary regulators with nonzero chirality mixing beginning beyond the linear jet, not another globally chirality-preserving onsite coin.

That escape resource is now exact at coefficient level. The quadratic regulator  $q(\mathbf{k})R$ , with  $q(\mathbf{k}) = \sum_j k_j^2$  and an explicit chirality-odd matrix  $R$ , vanishes at the Dirac point and has zero complete Fréchet derivative there, yet equals a nonzero chirality-mixing matrix on a unit-axis fixture. This is NEW **KERNEL**. It proves that higher-order mixing can preserve the desired first jet; it does not yet make the regulator an exact finite-range unitary or exclude every unwanted crossing.

The compatible exact-unitary primitive is a four-factor group commutator of circle phase steps. Kernel-checked, it is unitary, becomes the identity when either angle vanishes, gives  $+1$  for commuting involutions and  $-1$  for anticommuting involutions at a quarter turn, and admits an explicit rational mixed pair whose result is not any scalar matrix. Its algebraic Lie coefficient  $GA - AG$  is chirality-odd whenever  $A$  is chirality-even and  $G$  is chirality-odd; for the live  $(\alpha_1, \beta)$  pair that coefficient is nonzero and equals its full odd projection. The complete first Fréchet derivative of the commutator vanishes at the origin, while its mixed second derivative is exactly  $GA - AG$ . These are NEW **KERNEL**. The same exact algebra also proves a sharp negative result. A common sign flip of either cosine/sine pair leaves the commutator unchanged because the corresponding central signs occur twice and cancel. Hence every integer-frequency phase commutator, including one with affine phase offsets, takes the same value at all cubic  $0/\pi$  corners as at the origin; every finite product of zero-offset loops is exactly invisible there. A single phase step supplies the negative control and does see the sign flip. Thus the obstruction belongs to the doubled commutator architecture, not to phase gates individually.

The first exact primitive to survive that obstruction uses a genuinely noncentral conditional shift. With  $D(z) = \text{diag}(z, 1)$ , an exact rational coin  $C$ , and  $K(z) = D(z)CD(z^{-1})C^{-1}$ , define



the reciprocal word  $S(z) = K(z)K(z^{-1})$ . Lean proves that  $S$  is unitary on the circle,  $\det S = 1$ ,  $S(1) = \mathbf{1}$ , and  $S(z) - \mathbf{1} = (z - 1)^2 Q(z)$ . Thus its constant and linear jets vanish exactly. Its strict locality is also exact rather than inferred:  $S$  has five Laurent coefficients at powers  $-2, -1, 0, 1, 2$ , both range-two coefficients are nonzero, their zeroth moment is the identity, and their first moment vanishes. After multiplication by  $z^2$ , the word is an ordinary degree-four matrix polynomial. At  $z = -1$  the resulting rational matrix has both  $\det(S - \mathbf{1}) = 2304/625$  and  $\det(S + \mathbf{1}) = 196/625$ , so it removes both zero- and  $\pi$ -crossings of the identity baseline at that corner. This is NEW [KERNEL](#) as a two-band finite-Laurent construction primitive, not yet an alias-free  $3 + 1$  walk. A naive four-component embedding creates additional generic roots in the external oracle; any successful embedding now owes an exact all-torus root certificate.

The two-band primitive itself is now classified across the full rational tangent-half-angle coin family. Writing  $c = (1 - r^2)/(1 + r^2)$  and  $s = 2r/(1 + r^2)$ , the reciprocal word remains exactly unitary with determinant one on the circle. At  $z = -1$ , its zero-gap determinant is nonzero exactly when  $r \neq 0, \pm 1$ , while its  $\pi$ -gap determinant is nonzero exactly away from the roots of  $r^2 - 2r - 1$  and  $r^2 + 2r - 1$ . The original 3-4-5 coin is exactly the  $r = 1/2$  member, and  $r = 0$  is the identity negative control. These are NEW [KERNEL](#) two-band corner statements. They do not define the four-component blocks or provide a global-torus embedding certificate.

There is a precise gate on one tempting certificate. For a unitary  $4 \times 4$  step with  $\det U = 1$ , the determinant  $\det(U - \mathbf{1})$  is exactly real: the kernel-checked identity is

$$\overline{\det(U - \mathbf{1})} = \det(U - \mathbf{1}).$$

This makes a real intermediate-value argument available when two Weyl sectors have paired determinant. Conversely, real endpoint values alone do not make  $\det(U(q) - \mathbf{1})$  real along a path. The family-level reciprocal sign oracle therefore remains conditional until the live  $U_{\pm}(q)$  blocks, one consistent register convention, paired determinant along an explicit origin-avoiding path, and the endpoint formulas are composed in one theorem. This condition was exposed by hostile audit and is now NewResult Kernel as a reusable reality gate, not as a root-existence theorem.

At the two-band level there is a complementary exact locking theorem. If  $U$  is unitary and  $\det U = 1$ , then

$$\det(U - \mathbf{1}) = 0 \iff U = \mathbf{1}, \quad \det(U + \mathbf{1}) = 0 \iff U = -\mathbf{1}.$$

Thus determinant roots and whole-sector zero/ $\pi$  crossings coincide in an  $SU(2)$  block; they are not merely statements that one eigenvalue happened to hit the branch. Exact determinant-minus-one controls show the conclusion fails without unimodularity. This is NEW [KERNEL](#) and supplies the legal bridge between matrix-root and identity-preimage censuses used below.

A more serious escape attempt puts the reciprocal mixing inside a genuinely chirality-coupled four-component coin, so that the regulator is noncentral at the old corners and leaves the Dirac first jet only at quadratic order. That direct construction also fails. On the exact slice  $q_x = \pi$ ,  $q_y = 0$ , its zero- and  $\pi$ -crossing determinants reduce to squared reciprocal quartics. With  $x = z + z^{-1} = 2 \cos q_z$ , their reduced factors have opposite signs on  $x \in [1, 2]$  and  $x \in [-1, 0]$ , respectively. Hence the coupled word has at least one nonzero-momentum crossing of each quasienergy on that slice. Lean proves both physical unit-circle existentials exactly: the reduced real quadratics cross zero, the intermediate value theorem supplies roots, and  $q_z$  is reconstructed with arccos, with  $q_z \neq 0$  in both cases. This is NO-GO [KERNEL](#) for the displayed direct construction, not a universal reciprocal no-go. The negative result is diagnostic: corner visibility and higher-order chirality mixing can move the alias locus without reducing it. An enlarged determinant-paired register or an explicitly minimally doubled construction is the next honest test.

A second exact primitive shows that a stationary onsite amplitude need not conflict with strict locality or unitarity. For arbitrary orthogonal projectors  $P$  and  $Q$ , not assumed to commute, the controlled-phase product has the range-one Laurent form

$$U(z) = z\Gamma_+ + \Gamma_0 + z^{-1}\Gamma_-, \quad |z| = 1,$$

and is exactly unitary. A rational noncommuting two-band witness has  $\Gamma_0 = \begin{pmatrix} 16/25 & -12/25 \\ 12/25 & 16/25 \end{pmatrix} \neq 0$ . The necessary  $z \neq 0$  hypothesis and the correct same-family adjoint laws were enforced during formalization after the originally proposed formulas failed. This is NewResult Kernel as a one-variable construction mechanism inspired by the stationary-amplitude route of Gupta and Short. It does not yet supply a three-dimensional isotropic symbol, a Dirac tangent, or an all-zone proof that only the intended Weyl cones remain.

The first of those gaps is now closed constructively. Three explicit rational noncommuting projector pairs have first Laurent moments exactly  $(3/5)\sigma_x$ ,  $(3/5)\sigma_y$ , and  $(3/5)\sigma_z$ . Every axis retains a nonzero onsite amplitude, while their ordered three-axis product is exactly unitary for every torus momentum and equals the identity at the origin. Thus strict locality, stationary amplitude, all-zone unitarity, and an isotropic set of per-axis Weyl first moments are compatible in one NewResult Kernel construction. At the origin, the other two factors are identities, so the product-level isotropic first jet follows by the ordinary product rule; that combined derivative has not yet been packaged as a separate kernel theorem. The claim stops there: the full zero- and  $\pi$ -crossing census of this new symbol must determine whether it is minimally doubled or still more highly aliased.

The first global test is already negative. At the distinct torus phase  $(z_x, z_y, z_z) = (-1, 1, -1)$ , the complete ordered symbol is exactly the identity. Both the matrix equality and distinctness from  $(1, 1, 1)$  are NoGo Kernel. Thus the explicit stationary-amplitude construction is not unique-cone. This is not the hollow case of a constant identity symbol: at  $(z_x, z_y, z_z) = (i, 1, 1)$  its  $(0, 0)$  entry is exactly  $16/25$ , so the symbol is kernel-proved nonconstant. An exact-coefficient oracle consistently finds two additional off-corner identity roots. The first is now an exact kernel theorem:

$$(z_x, z_y, z_z) = \left( \frac{-9 - 40i}{41}, \frac{-9 + 40i}{41}, 1 \right), \quad U(z_x, z_y, z_z) = I.$$

Both nontrivial phases lie exactly on the unit circle and differ from 1. Conjugate pairing is load-bearing: replacing  $z_y$  by  $z_x$  makes the  $(0, 1)$  matrix entry exactly  $-1104/1681$ , so the resulting symbol is not the identity. This off-corner crossing is NoGo Kernel. A second, fully off-axis crossing is now exact as well. In tangent coordinates its third coordinate is the unique real root of

$$480t^5 - 575t^4 - 1026t^2 + 1440t - 575 = 0, \quad 149/100 < t < 3/2.$$

Explicit rational quartic functions of  $t$  reconstruct the other two tangent coordinates. Lean proves all three are nonzero, maps them to unit-circle phases different from 1, and proves that the actual imported live matrix is exactly  $U = I$ . The companion elimination sextic is strictly positive on all of  $\mathbb{R}$ . The corrected original-generator certificate is now exact in Lean as well:

$$(1 + t_z^2)^2 t_z p_5(t_z) p_6(t_z) = Q_x F_x + Q_y F_y + Q_z F_z.$$

The three quotient polynomials contain 522 exact rational terms in total. Kernel normalization proves this identity and, because  $1 + t_z^2 > 0$ , proves that simultaneous numerator zeros force the zero, quintic, or sextic branch. The chart factor is mandatory: the corresponding bare polynomial is not in the numerator ideal. These are NEW **KERNEL**. The formerly missing live bridge is now exact: on the finite real tangent chart, Lean proves that the actual imported matrix satisfies  $U = I$  if and only



if its scalar equation and all three numerator equations hold. On the full  $t_z = 0$  slice, those three live numerator equations have exactly two real solutions, the origin and  $(-5/4, 5/4, 0)$ ; both reconstruct  $U = I$ , and the latter is precisely the 9–40–41 alias. The interior census is now complete: exact polynomial-ideal certificates force  $t_x$  and  $t_y$  to be the displayed quartic reconstruction functions of  $t_z$  on the unique real quintic branch. Hence every finite-chart identity crossing is exactly the origin, the 9–40–41 point, or the unique fibre over that quintic root; the last branch has a fully off-axis witness. **NEW KERNEL** An exact boundary oracle finds only the already known  $(-1, 1, -1)$  identity crossing on the seven phase- $-1$  chart strata, but that boundary classification remains a separate formal gate. The result is therefore an exhaustive finite-chart no-go with a boundary-chart obligation, not a finished regulator.

A finite  $\mathbb{Z}_2^3$  cover can fold the eight  $\{0, \pi\}^3$  corner labels to one reduced representative while retaining a unique three-bit flavour label. The cover is bijective, its deck action is free and transitive, and a single diagonal  $\mathbb{Z}_2$  bit is provably too small to label all corners. This is **NewResult Kernel** as an exact census, but not a cure: it relabels eight crossings and removes none. For the present successive-axis symbol the exact translation intertwiner reduces to a scalar parity sign. Four flavour labels preserve zero quasienergy and four exchange zero with pi; a one-axis translation is explicitly nonidentity, while a  $\pi/2$  shift fails to be any scalar deck action. These statements are **NewResult Kernel**. This route must therefore not be advertised as a unique-cone theorem or as a derivation of observed particle generations: it is a faithful eightfold census.

The finite logical hinge is also kernel-checked: if a finite crossing set has total integer Jacobian-sign charge zero and one crossing is nondegenerate, a distinct nondegenerate crossing must exist. The theorem makes the topological input visible rather than hiding it: the zero-total premise still has to come from an exact crossing census or the separately audited Laurent/Floquet composition.

There is already a constructive Hamiltonian-level route around this no-go. Using the same live Clifford generators, define

$$M_W(\mathbf{q}) = m + r \sum_{j=1}^3 (1 - \cos q_j), \quad (60)$$

$$H_W(\mathbf{q}) = \sum_{j=1}^3 \sin q_j \alpha_j + M_W(\mathbf{q}) \beta. \quad (61)$$

The kernel-checked Wilson identity is

$$H_W(\mathbf{q})^2 = \left( \sum_{j=1}^3 \sin^2 q_j + M_W(\mathbf{q})^2 \right) \mathbf{1}. \quad (62)$$

For  $m > 0$  and  $r \geq 0$ , the scalar coefficient is bounded below by  $m^2$ . For  $m = 0$  and  $r > 0$ , it vanishes if and only if  $\cos q_1 = \cos q_2 = \cos q_3 = 1$ ; hence all non-origin cubic corners are lifted. For example, the  $(\pi, 0, 0)$  energy squared is exactly  $4r^2$ . This closes the spectral design problem at the nearest-neighbor Hamiltonian level while preserving the Dirac tangent. It does *not* yet close the stricter QCA problem:  $\exp(-itH_W)$  is exactly unitary, but its finite-time exponential is not asserted to be a finite Laurent polynomial or a strictly finite-range one-step update.

**Exact temporal blocking and nonclosure.** Define the two-step blocking map by  $\mathcal{B}(U) = U^2$ . The mass-only Pluecker subgroup closes exactly,

$$\mathcal{B}(C_z(\varepsilon)) = C_z(2\varepsilon),$$

but the ordered spatial split family does not close by doubling its scalar step. At the exact  $x$ - $y$  quarter-turn witness,

$$\mathcal{B}(U_{xy}(\pi/2)) = -\mathbf{1}, \quad U_{xy}(\pi) = \mathbf{1}. \quad (63)$$

The smallest multiplicatively closed enlargement is the submonoid generated by all bare split steps. These statements are **NEW KERNEL**: temporal blocking generates ordered word/coupling data rather than merely a renormalized copy of the original scalar parameters. Spatial decimation is a separate, stronger RG problem.

**A proved boundary on scale selection.** For the finite positive Pluecker action developed here, simultaneous scaling of both primitive spinors by  $t$  gives the exact law

$$S(t\psi, t\phi; x) = t^4 S(\psi, \phi; x). \quad (64)$$

Whenever the unscaled action is positive, its radial equation  $4t^3 S(\psi, \phi; x) = 0$  is equivalent to  $t = 0$ . The orthogonal-spinor, positive-quartet control has  $S = 1/2$ , so the obstruction is nonvacuous. The formal result is now more general. For any positive finite action with radial profile  $S(tx) = t^p S(x)$  and natural degree  $p > 0$ , the derivative at unit scale is  $pS(x) > 0$ , so the configuration cannot be stationary. At  $p = 0$  the radial direction is exactly flat and cannot select an isolated scale. This is **NO-GO KERNEL**: no positive fixed finite homogeneous action can spontaneously select a nonzero scale. A running/refining family, competing homogeneity, constraint, ensemble scale, or other dimensionful input must be displayed. The exact positive control  $V_c(t) = (t^2 - c)^2$  has stationary points precisely at  $t = 0$  or  $t^2 = c$ ; for supplied  $c = 1$ , the nonzero points  $t = \pm 1$  are global minima. Selection is therefore possible, but its magnitude is exactly the new input. This control is also **REALIZATION KERNEL**.

**Pre-registered high-momentum benchmark.** Before evaluation we fixed  $z = 3 + 4i$ ,  $\varepsilon = 0.2$ , Wilson  $r = 1$ , and a  $10^{-10}$  kill threshold. The oracle then found exactly three non-origin even-corner aliases (maximum residual  $2.30 \times 10^{-16}$ ), both body-center  $\pm 1$  modes (residuals below  $2.24 \times 10^{-16}$ ), no non-origin Wilson zero, and minimum Wilson corner gap 2. With  $r = 0$ , all eight corners became gapless as required by the negative control. No parameter was fitted. This numerical result validates the implementation of the theorem chain; the Lean identities, not the benchmark, carry the proof.

**Comparison with the tetrahedral walk.** Nzongani et al. [11] place four amplitudes on tetrahedral facets, compose facet/strict/weakly local shifts with directional coins, and insert the assigned coin  $e^{-i\varepsilon m \gamma^0}$ ; their expansion recovers the  $3 + 1$  Dirac equation to first order. Our register and finite update are different: the present construction is a cubic successive-axis product with a Pluecker-derived complex mass coin and an explicit many-step rate. A constant onsite basis change cannot decide away this difference: unitary conjugacy preserves equality of symbols at distinct momenta, so every constant onsite conjugate of our walk inherits the three aliases above and the body-center  $\{+1, -1\}$  spectrum. Any equivalence with a tetrahedral update must therefore reproduce those obstructions or use a non-onsite blocking, changed unit cell, or coarse-graining map. A full spectral audit of the tetrahedral symbol is needed before choosing among those possibilities.

**Small-step-sensitive  $3 + 1$  rate.** The live four-factor symbol now has a sharper fixed-time theorem. With  $B_4 = |k_x| + |k_y| + |k_z| + |m|$  and  $n > 0$ ,

$$\|U(t/n)^n - e^{-itH}\| \leq \frac{2B_4^2 t^2}{n} \exp\left(\frac{|t|B_4}{n}\right).$$

### High-symmetry audit of the finite 3+1 split walk

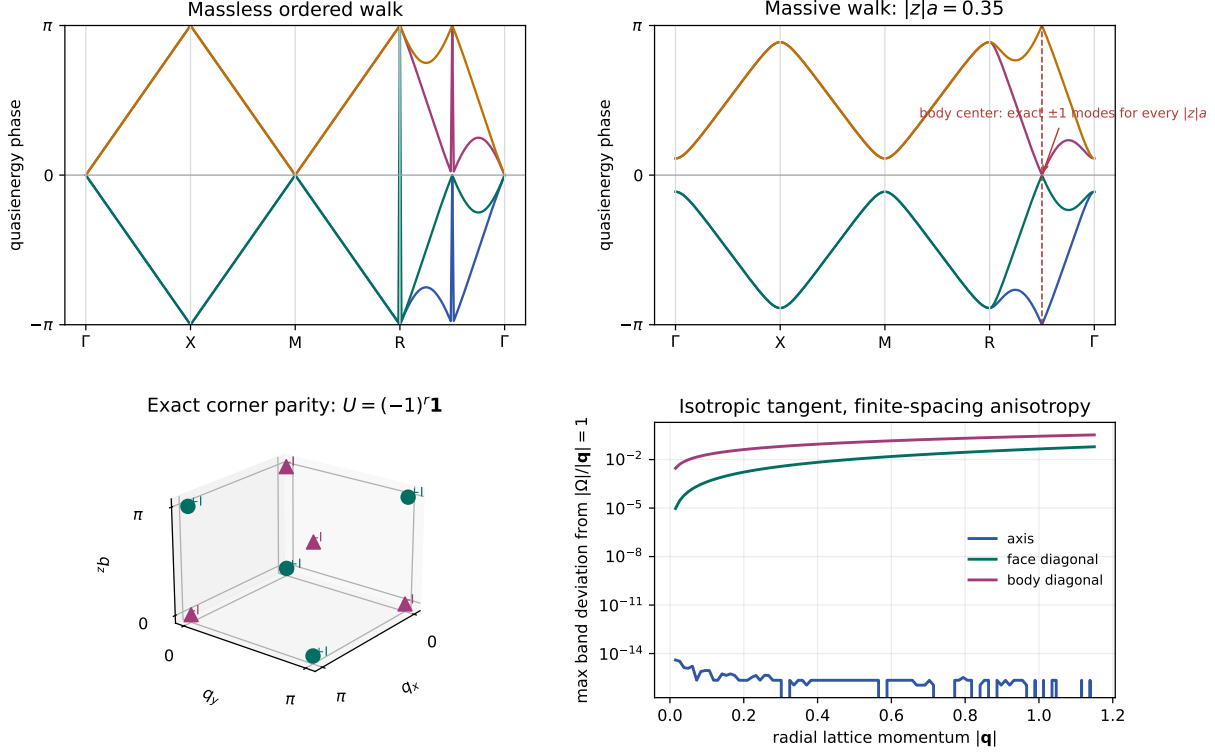


Figure 4: High-symmetry audit of the ordered 3 + 1 walk. The small-momentum tangent is radially isotropic, but the exact massless lattice symbol has four zero-quasienergy corners: the origin and three even-parity aliases. The marked body-center crossing in the massive panel persists for every mass angle by Eq. (46). Curves evaluate the exact finite matrices; the corner classification and body-center eigenmodes are kernel checked.

The proof keeps the small-step exponent through both the split-product remainder and the unitary telescope; unlike the earlier compact-box envelope, it does not replace this factor by  $\exp(B_4)$ . The resulting growing-window theorem takes  $n = K^4$  and permits arbitrary momentum and mass sequences with  $B_4 \leq K$ ; their modewise split-versus-exact error is bounded by  $2t^2 K^{-2} \exp(|t|K^{-3})$  and tends to zero. A sharp pure-momentum boundary fixture saturates  $B_4 = K$ , so this is not a fixed-mode restatement. Shannon sampling/interpolation into physical space and the position-space  $L^2$  Dirac PDE limit remain Paper D.

## 8 Formal verification and negative controls

The formal development uses Lean 4.28.0 and Mathlib [20, 21]. Its spinor and physics conventions were also cross-checked against the public PhysLean corpus [22]. Formalized mathematical physics at substantial scale is becoming practical: the free bosonic quantum field theory in four-dimensional Euclidean spacetime, satisfying the Glimm–Jaffe axioms, was recently constructed end to end in Lean 4 [23]. The present artifact is complementary in scope: exact finite discrete dynamics, with the physical input — the rest operator itself — derived inside the formalization rather than axiomatized. The trusted Cauchy–Binet identity lives in the project core; the newer

operator and walk results remain in the draft namespace while their publication interfaces are reviewed. Each headline declaration below has a build-enforced assumption footprint contained in `{propext, Classical.choice, Quot.sound}`.

| Declaration                                                       | Status                | Mathematical payload                                                                                                                                          |
|-------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| two_edge_plucker_mass_identity                                    | CLASSICAL<br>KERNEL   | $\det P =  \psi \wedge \phi ^2$                                                                                                                               |
| fin_bundle_plucker_mass_identity                                  | CLASSICAL<br>KERNEL   | finite-family Cauchy–Binet area sum                                                                                                                           |
| hermitian_determinant_uniqueness                                  | CLASSICAL<br>KERNEL   | determinant is the unique quadratic null-boundary form up to scale                                                                                            |
| pair_diracSymbol_sq                                               | NEW KERNEL            | $H_z(k)^2 = (k^2 + \det P)\mathbf{1}$ for arbitrary pairs                                                                                                     |
| massOperator_eq_zero_iff_collinear                                | NEW KERNEL            | exact gap-closing locus                                                                                                                                       |
| phase_covariance                                                  | NEW KERNEL            | oriented phase acts by conjugacy                                                                                                                              |
| massOperator_sl2_invariant                                        | NEW KERNEL            | invariance under the defining $\mathrm{SL}(2, \mathbb{C})$ action                                                                                             |
| oddHermitian_eq_massOperator                                      | NEW KERNEL            | complete normal form for odd Hermitian rest operators                                                                                                         |
| massCoin_unitary_group                                            | NEW KERNEL            | exact unitary rest evolution and group law                                                                                                                    |
| sameMomentum_covariant_iff,<br>parity_covariant_iff               | NEW KERNEL            | complete covariance classification of the momentum-dependent two-channel generator: diagonal phases at fixed momentum and antidiagonal flips only with parity |
| chiralPhase_walk_covariance,<br>chiralFlip_walk_parity_covariance | NEW KERNEL            | both classified branches act exactly on the ordered finite step $T(k)C(z)$ ; exhaustive discrete-step symmetry classification is not claimed                  |
| unitary_kernel_eq_scaled_checkboard                               | REALIZATION<br>KERNEL | exact $\cos^n$ scaling of the two-weight recursion; operator/history identification is supplied by separate transfer-matrix theorems                          |
| exact_quantum_walk_dispersion                                     | REALIZATION<br>KERNEL | concrete trace identity, trace-to-quasienergy bridge, unitarity, determinant one, and rest eigenphases                                                        |
| massive_implies_subluminal                                        | REALIZATION<br>KERNEL | strict full-zone velocity-deficit inequality for massive angle; derivative identity is <a href="#">PROOF</a>                                                  |
| massless_gapless_classification                                   | REALIZATION<br>KERNEL | exact $\{0, \pi\}$ massless band-touching set                                                                                                                 |
| pathAmplitude_eq_corner_power                                     | REALIZATION<br>KERNEL | exact real-mass Gaussian-rational corner-power weight; complex orientation is carried by phase conjugacy                                                      |
| complexPlueckerPathSum_eq_transfer_power                          | NEW KERNEL            | direct directed-edge history sum using the actual complex coin entries $z$ and $\bar{z}$                                                                      |
| fixed_time_many_step_bound                                        | NEW KERNEL            | fixed-parameter $D(k, \mu)t^2/n$ estimate                                                                                                                     |
| fixed_time_many_step_bound_on_norm                                | NEW KERNEL            | uniform bounded-parameter estimate                                                                                                                            |
| localStep_preserves_norm                                          | REALIZATION<br>KERNEL | exact finite-torus $3 + 1$ norm preservation                                                                                                                  |
| inverseSuccessiveWalk_after,<br>successiveWalk_after_inverse      | REALIZATION<br>KERNEL | explicit two-sided inverse of the complete local cycle                                                                                                        |
| successiveWalk_zero_of_causal_source_zero                         | REALIZATION<br>KERNEL | strict one-shift-per-axis causal cone                                                                                                                         |
| Finite3Plus1FourierBridge.localStep_modes                         | REALIZATION<br>KERNEL | exact finite-character block on every product plane wave                                                                                                      |
| finiteLocalSymbol_eq_analytic_deg                                 | REALIZATION<br>KERNEL | exact finite-to-analytic momentum-sign conversion                                                                                                             |
| quarter_zone_sign_control                                         | REALIZATION<br>KERNEL | nondegenerate $-i$ convention fixture                                                                                                                         |
| Pluecker3Plus1ComplexMass.H4_sq                                   | NEW KERNEL            | phase-retaining $3 + 1$ mass gives $H_4^2 = ( \mathbf{k} ^2 +  z ^2)\mathbf{1}$                                                                               |
| complex_phase_covariance                                          | NEW KERNEL            | chiral covariance of the complex four-component rest operator                                                                                                 |
| massCoin4_unitary_group                                           | NEW KERNEL            | derived complex mass coin is unitary with exact group law                                                                                                     |
| variableComplexLocalStep_preserves_norm                           | NEW KERNEL            | arbitrary position-dependent Pluecker profiles, including zero sites, evolve unitarily                                                                        |
| variableComplexLocalStep_zero_of_causal_source_zero               | NEW KERNEL            | strict one-shift-per-axis causal cone for the inhomogeneous complex-Pluecker walk                                                                             |
| local_phase_connection_covariance                                 | NEW KERNEL            | site-dependent phase induces exact endpoint links and rotates $z(x)$ pointwise                                                                                |
| gaugedCliffordAxisShift_constant                                  | NEW KERNEL            | constant phase produces no endpoint link                                                                                                                      |
| global_real_lift_forces_zero_winding                              | NEW KERNEL            | a global real phase lift has zero winding; nonzero index needs branch/transition data                                                                         |

The controls are part of the result rather than decorative examples.

| Claim                      | Positive/nondegenerate control                       | Failure or boundary control                                                           |
|----------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------|
| Area creates a gap         | standard basis pair gives $z = 1$ and $B_z^2 = 1$    | collinear pair $(1, 2), (3, 6)$ gives $z = 0$ and $B_z = 0$                           |
| Full null boundary matters | determinant vanishes on every rank-one edge          | the form $ac$ vanishes on coordinate rays but fails on the diagonal ray               |
| Mass enters at turns       | a three-step history is nonzero and mass-dependent   | at zero mass every positive-corner history vanishes                                   |
| Many-step convergence      | rational $(k, \mu) = (3/5, 4/5)$ satisfies the bound | reversing the mass phase converges to the wrong Hamiltonian numerically               |
| 3 + 1 factor composition   | $(1, 2, 2, 3)$ gives $H_4^2 = 181$ and nonzero norm  | $\alpha_1\alpha_2 \neq \alpha_2\alpha_1$ forbids order erasure and naive second order |

The principal verification commands are

```
lake env lean PhysicsSM/Draft/NullEdge/PluckerMassOperator.lean
lake env lean PhysicsSM/Draft/NullEdge/PluckerMassDynamics.lean
lake env lean PhysicsSM/Draft/NullEdge/CornerWalkEquivalence.lean
lake env lean PhysicsSM/Draft/NullEdge/FermionDoublingAudit.lean
lake env lean PhysicsSM/Draft/NullEdge/ExactCheckerboardPathSum.lean
lake build PhysicsSM.Draft.NullEdge.ExactQuantumWalkDispersion
lake build PhysicsSM.Draft.NullEdge.FixedMomentumManyStepContinuum
lake build PhysicsSM.Draft.NullEdge.BoundedMomentumManyStepContinuum
lake build PhysicsSM.Draft.NullEdge.Compact3Plus1DiracRate
lake env lean PhysicsSM/Draft/NullEdge/Compact3Plus1RefinedWindowRate.lean
lake env lean PhysicsSM/Draft/NullEdge/Local3Plus1RateBridge.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteWalkPositionConvergence.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteZModPlancherel.lean
lake env lean PhysicsSM/Draft/NullEdge/Finite3Plus1FourierBridge.lean
lake env lean PhysicsSM/Draft/NullEdge/Finite3Plus1AnalyticSignBridge.lean
lake env lean PhysicsSM/Draft/NullEdge/Pluecker3Plus1ComplexMass.lean
lake env lean PhysicsSM/Draft/NullEdge/ComplexPlueckerLocalWalk.lean
lake env lean PhysicsSM/Draft/NullEdge/ComplexPlueckerRateTransfer.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteTorus3Plancherel.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteTorus3WalkWavepacket.lean
lake env lean PhysicsSM/Draft/NullEdge/Finite3Plus1BrillouinAudit.lean
lake env lean PhysicsSM/Draft/NullEdge/StationaryAmplitudeNoGo.lean
lake env lean PhysicsSM/Draft/NullEdge/StationaryAmplitudeLiveAxisNoGo.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteWalkOnsiteEquivalenceObstruction.lean
lake env lean PhysicsSM/Draft/NullEdge/LocalQCAProperties.lean
lake env lean PhysicsSM/Draft/NullEdge/CountableL2WavepacketConvergence.lean
lake env lean PhysicsSM/Draft/NullEdge/ContinuumL2MultiplierBridge.lean
lake env lean PhysicsSM/Draft/NullEdge/CompactSupportL2WalkBridge.lean
lake env lean PhysicsSM/Draft/NullEdge/VariablePlueckerLocalWalk.lean
lake env lean PhysicsSM/Draft/NullEdge/VariablePlueckerPhaseConnection.lean
lake env lean PhysicsSM/Draft/NullEdge/ComplexPlueckerCheckerboardPathSum.lean
lake env lean PhysicsSM/Draft/NullEdge/GlobalPhaseWindingNoGo.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerWindingDerived.lean
lake env lean PhysicsSM/Draft/NullEdge/StrictQCAMinimalArchitecture.lean
```

```

lake env lean PhysicsSM/Draft/NullEdge/CommutatorWilsonStrictnessKill.lean
lake env lean PhysicsSM/Draft/NullEdge/TemporalBlockingRG.lean
lake env lean PhysicsSM/Draft/NullEdge/FullBlochSplitPlus.lean
lake env lean PhysicsSM/Draft/NullEdge/FullBlochSplitMinus.lean
lake env lean PhysicsSM/Draft/NullEdge/FullBlochZeroClassification.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteCARFockBasic.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteCARSecondQuantization.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerQuarticInteraction.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerCausalCone.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerLayerCone.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerPairGenerator.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerRestOperatorGeneral.lean
lake env lean PhysicsSM/Draft/NullEdge/PairKickSelection.lean
lake env lean PhysicsSM/Draft/NullEdge/SplitStepChargeBalance.lean
lake env lean PhysicsSM/Draft/NullEdge/MassCovarianceForcing.lean
lake env lean PhysicsSM/Draft/NullEdge/CARBlockReduction.lean
lake env lean PhysicsSM/Draft/NullEdge/SplitStepSchurJet.lean
lake env lean PhysicsSM/Draft/NullEdge/SplitStepSchurJetAllNodes.lean
lake env lean PhysicsSM/Draft/NullEdge/TwoBandCrossingDoubling.lean
lake env lean PhysicsSM/Draft/NullEdge/TwoBandFlowCount.lean
lake env lean PhysicsSM/Draft/NullEdge/TwoBandEigenphaseAnalytic.lean
lake env lean PhysicsSM/Draft/NullEdge/Selection2Repairs.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerPhaseObservable.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerPhaseDefectSpectrum.lean
lake env lean PhysicsSM/Draft/NullEdge/ChangingLatticePDECORE.lean
lake env lean PhysicsSM/Draft/NullEdge/Carrier/PluckerScaleSelectionNoGo.lean
lake build PhysicsSM.Draft.NullEdge.OvernightTheoryAxiomGuard
python Scripts/sim/null_edge_regulator_benchmark.py
python Scripts/sim/plot_null_spinor_dirac_bands.py
python Scripts/sim/plot_null_edge_paper_figures.py
python Scripts/publication/verify_null_edge_paper_a.py

```

The cited modules contain no proof handoff markers or compiled-evaluator proofs. The working tree is not yet an archival release. A clean commit, source archive, theorem ledger, and DOI remain submission requirements.

## What mechanization changed

The formal artifact is scientifically useful only where it improves the mathematics. In this development it forced ten corrections that are visible in the paper. First, it separated the first-order rest energy  $\mu$  from the quadratic invariant  $\mu^2 = \det P$  instead of allowing one symbol to drift between operator orders. Second, exact unitarity exposed the finite-spacing corner ratio  $\tan(a\mu)$ , replacing the tempting but false finite identity  $a\mu = \tan(a\mu)$ . Third, proving an iterated product estimate separated a one-step Taylor expansion from genuine uniform many-step control. These are not the only bookkeeping gains. Fourth, the exact  $3 + 1$  composition preserved the noncommuting  $x/y/z$ /mass factor order instead of identifying reversed local updates. Fifth, the  $1 + 1$  full-zone audit separated the zero-quasienergy cone from its high-quasienergy Floquet partner. Sixth, the  $3 + 1$  high-symmetry audit disproved the tempting global reading of continuum isotropy: the tangent is isotropic, three nonzero corners exactly alias the origin, and at body center the massive step retains

explicit  $+1$  and  $-1$  modes for every mass angle. Seventh, the exact Laurent residual ruled out the most economical proposed repair: a nonzero stationary amplitude cannot coexist with origin normalization, exact unitarity, and a full involutory tangent inside one degree-one nearest-neighbor factor. Eighth, the inhomogeneous phase calculation separated a global basis rotation from a genuine endpoint connection and proved that a global real lift cannot carry nonzero winding. Ninth, exact temporal blocking showed that the mass-only group law closes while the noncommuting spatial split generates a larger ordered-word family. Tenth, the radial derivative of the positive Pluecker action proved that its homogeneity collapses rather than selects a nonzero scale. These are the formalization contributions; the number of declarations is not.

## 9 What is forced, derived, realized, and open

The central scientific distinction is not between bold and cautious language. It is between different logical verbs.

| Verb     | Claim in this paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forced   | A quadratic invariant vanishing on the full null boundary is proportional to the determinant; a nonnegative first-order gap is proportional to its square root.                                                                                                                                                                                                                                                                                                                    |
| Derived  | The spinor pair gives $z$ ; $z$ gives $B_z$ ; $B_z$ gives the Dirac square, rest gap, unitary turn gate, and orientation-sensitive corner phase; a site field $z(x)$ gives its own patched link data with an exact $2\pi$ -integer winding.                                                                                                                                                                                                                                        |
| Realized | The split-step null walk exhibits the construction exactly and converges uniformly on bounded $1+1$ momentum boxes; a finite local $3+1$ successive-axis walk has an ordered complex-Pluecker symbol with the same explicit rate, a proved quartically growing-window consistency theorem, exact three-torus Plancherel control, and a classified high-symmetry failure set.                                                                                                       |
| Open     | A walk-specific changing-lattice Fourier/PDE limit, a replacement $3+1$ regulator that removes the corner aliases and body-center modes, beyond the excluded degree-one stationary-amplitude ansatz, reduction of the derived link data to a corrected non-winding-only defect index and stable localized mode, instantiation of the proved general free-plus-pair CAR cone with the full live $3+1$ walk, an operator-exponential/action law, and a continuum interaction theory. |

What can an assigned scalar mass mimic? Everything that depends on  $|z|$  alone: after setting  $m = |z|$ , the constant-coefficient spectrum, the dispersion, the velocity bound, and the uniform many-step limit are identical, and the paper says so without embarrassment. What it cannot mimic is now a theorem list rather than a narrative. It has no input from which the rest operator is derived rather than declared (the classification theorem makes “derived” precise); it carries one real coin where the exact history sum carries  $z$  and  $\bar{z}$  on opposite orientations; it produces no endpoint connection, no integer link winding, and no patched transition data from a varying phase unless that field is separately supplied; and a scalar mass, being one real number, cannot reproduce the exact  $4/5$ -versus-1 interference split in the supplied pair-kick sector. More importantly, it cannot reproduce the free two-site spectral split between equal-modulus fields without importing exactly the complex datum and its parallel transport that this paper derives. A spatially common constant phase remains removable, as it must; a transported relative phase is already a free-spectrum observable. The pair interaction is a second phase reader, not the condition that makes the free



reader exist. Once the phase varies around a cycle, the distinction further becomes patched link data with an integer winding.

There is now also a free two-site spectral consequence. For two equal-modulus rest blocks coupled by the displayed unit edge phase, the exact  $4 \times 4$  Hamiltonian obeys a scalar quadratic polynomial whose splitting is controlled by the parallel-transported mismatch  $\Delta = z_R - (\bar{w})^2 z_L$ . Its lower squared-energy level vanishes exactly at  $t = |z_L|$  and  $z_R = -(\bar{w})^2 z_L$  (for  $t \geq 0$ ), while a common phase on both sites is removed by an explicit unitary conjugacy. The equal-modulus hypothesis is load-bearing: the formal development contains a concrete unequal-modulus counterexample. This is a phase-sensitive finite free spectrum, not yet a topological-protection or localization theorem.

The paper proves a finite free construction and one finite phase-sensitive quartic extension. It does not prove that every massive particle is composite in the ordinary particle-physics sense. The “constituents” here are null momentum factors of a finite spinor description, not additional asymptotic particles. Nor does the theorem compute an absolute mass scale: rescaling the input spinors rescales  $z$  and hence  $\mu$ . The quartic-action no-go sharpens this from an omission to a proved boundary for the natural positive homogeneous action.

The result does support a precise statement: a timelike rest gap can be encoded entirely by massless rank-one momentum primitives, and the gap is exactly the area opened between their spinor directions. In the derived walk, microscopic translation occurs along null channels while subluminal drift and a rest gap arise from coherent direction changes. This is an exact property of the finite model, not yet a universal ontology of nature.

## 10 Interpretation and open problems

The larger null-information program studies a four-term finite carrier square, positive cohomology, closure interactions, and discrete frame data. Those structures motivated the present theorem but are not required to prove it. They belong in a companion paper unless they acquire a theorem that depends on  $B_z$ .

Three claim-discipline boundaries govern everything in this paper and its companions. First, every invariant here is a finite-dimensional algebraic discriminator on an explicit register; we claim no Fredholm or finite-volume topological index, and any comparison with established real-space walk indices is made through their closed-form evaluation on infinite periodic extensions, never as a finite-volume theorem. Second, the pair interaction is supplied as an even Hermitian four-mode kick consistent with the CAR support and layer schedule; we prove its exact gate, unitarity, and composition behavior, but we do not derive the interaction from a microscopic principle, and the same gap is open in the interacting-automaton literature we compare against. Third, every result in this paper is machine-checked in Lean 4 with its axiom footprint pinned; where companion analyses use exact computer-algebra computations that are not yet formalized, they are tagged as such there and are never cited here as theorems.

The next problems are sharply ordered.

1. **Complete the lattice-to-PDE identification.** The finite product DFT, its exact  $N^{-3}$  normalization, and the walk-specific compact-support momentum-space  $L^2$  limit are closed. Control the changing lattice/continuum Fourier spaces and identify the limiting multiplier with the position-space Dirac PDE propagator. The abstract bulk/tail split and vanishing of the ultraviolet tail under measurable monotone band exhaustion are now closed for arbitrary  $L^2$  data, and the per-step symbol estimate now carries the refined form  $2B_4^2 t^2 / n \cdot \exp(|t|B_4/n)$ , which retains the small-step factor that the earlier bound discarded; the remaining frontier is the balancing of a growing momentum window against the Sobolev tail, composed with the

concrete scaled sampling/interpolation isometry. Equations (33), (34), (35), and (36) close the finite, abstract countable, and compact-support multiplier steps, not that final physical identification.

2. **Remove the lattice partners without losing the derivation.** The no-go (53) rules out the naive degree-one stationary-amplitude factor with a full Dirac tangent. The Wilson construction (62) now removes the unwanted corners and gives a uniform massive gap for a nearest-neighbor Hamiltonian. The open task is to retain that spectral success in a strictly finite-range, exactly unitary discrete-time update. Test the enlarged internal cells, longer Laurent range, and tilted-projector stay-put constructions of the kind realized by Gupta and Short [16] while preserving the Pluecker-derived coin and exact path dictionary; by the corollary above, any such stay-put route must quantify what replaces the involutory unit-speed tangent. In  $3 + 1$ , remove the three proven even-parity origin aliases and both mass-independent body-center modes while retaining locality, unitarity, continuum isotropy, and the complex-mass rate, and eliminate the residual non-Dirac states that persist in the Gupta–Short family. The corrected search should separate a globally chirality-preserving control class from an escape class with a nonzero  $\Xi$ -odd term whose constant and linear jets vanish; exact group-commutator loops are the first such candidate. Equations (45) and (46) are the regression tests.
3. **Complete the tetrahedral comparison.** Compute the tetrahedral walk’s full Bloch symbol in a common blocked unit cell and audit its corners, bands, and locality radius. Constant onsite equivalence is already constrained by the alias-preservation no-go; a genuine comparison must specify any blocking or coarse-graining map.
4. **Generalize the Pluecker rest operator.** Now largely resolved in kernel-checked form (`PlueckerRestOperatorGeneral`): for *any* number of spinors, the pairwise-area matrix  $Z$  obeys the exact cube closure  $ZZ^\dagger Z = \mu^2 Z$  with  $\mu^2$  the complete area budget (Gram determinant; Lagrange identity  $\sum_{i < j} |z_{ij}|^2 = \mu^2$ ), the odd Hermitian block operator obeys  $B_w^3 = \mu^2 B_w$  — the same finite closure law as the interaction generator’s  $K^3 = |z|^2 K$  — with  $B_w^2$  equal to  $\mu^2$  times a rank-four support projector for every  $n$ , the two-spinor case recovering  $B_z^2 = \mu^2 \mathbf{1}$  on the whole space, the massless boundary  $Z = 0 \iff \mu^2 = 0$ , and an explicit non-decomposable antisymmetric control violating the cube law, so the spinor (Pluecker-decomposable) origin of the area data is load-bearing. What remains open is the decomposition-independent phase/moduli classification of  $B_w$  beyond the two-spinor case.
5. **Reduce the derived winding to an index and a localized mode.** The position-dependent unitary update, its inverse and causal cone, and exact site-dependent chiral connection are closed, and the derivation gap is now closed as well: the link increments of any nowhere-zero Pluecker field are the principal arguments of its neighbor ratios, their cycle sum is an exact  $2\pi$ -integer winding, a globally lifted real phase still forces zero, and an explicit winding-one field is generated by primitive spinors. The companion defect paper has now resolved the fixture-size version for sign/domain walls through the collinear zero set: a machine-checked positional law (chart self-adjointness holds exactly for the twelve of sixteen fields whose lone flip avoids a reflection-fixed site, as an *iff* across the whole coin continuum), protected exact  $\pm 1$  eigenvectors through an involutive-compression engine, a machine-checked multiplicity census (dimension 2 for singleton fields, 4 for domain blocks, 0 for controls), and a kernel-checked impossibility: no translation-invariant index of the periodic extension, composed with any decoder, reproduces the positional certificate. What remains open here is the general-length reduction: a finite

index operator for the *derived winding* on arbitrary registers, with a localized protected mode and a displayed stability class, and a theorem connecting it to the companion's positional data.

6. **Complete the many-body and interacting lift.** The generic finite CAR algebra, a phase-sensitive four-mode quartic pair-transfer observable, and an exact norm-preserving pair kick are now closed; the quartic operator is Hermitian and the kick/operator restriction identity is exact. The determinant-minor lift now fixes vacuum, agrees with the one-particle matrix, preserves number and parity, obeys exact creation covariance, and satisfies  $\Gamma(1) = 1$  and  $\Gamma(UV) = \Gamma(U)\Gamma(V)$ . Its ordered-minor adjoint law, lifted two-sided inverse, and finite Fock inner-product preservation are also exact. Annihilation covariance is now exact, and the placed quartic pair layer has exact disjoint-block commutation, finite scheduled CAR-support propagation, a sequential finite-range graph-neighborhood cone, and one neighborhood expansion per pairwise-disjoint circuit layer. The abstract free/interacting composition is now closed: a coefficient-local unitary one-particle update, lifted by determinant minors, contributes one neighborhood step and one disjoint pair layer contributes a second. Instantiate this theorem with the full live spatial walk, then derive an operator-exponential law and the quartic dynamics from one field-valued action whose reductions also produce the null-spinor geometry and one-particle evolution. The phase-reading selection problem is resolved at block level. Under the site-local chiral action  $\text{diag}(u, 1)$ , equivariance, Hermiticity for every field value, and vanishing at  $z = 0$  force  $H(z) = \begin{pmatrix} 0 & \alpha z \\ \alpha z & 0 \end{pmatrix}$ ; the converse and explicit hypothesis-violating controls are kernel theorems (`PairKickSelection`, `Selection2Repairs`). A scalar common phase would instead force  $H = 0$ , so the chiral weight is load-bearing.

This is a Fock-level result, not merely a chosen  $2 \times 2$  model. The pair-block Fock operators are linearly equivalent to  $M_2(\mathbb{C})$  with Hermiticity preserved, and the block gauge is the shadow of an explicit Fock unitary (`CARBlockReduction`). Moreover, the allowed chiral action is forced by the complete covariance group of the static family  $\{B_z\}$ , which is the chiral phase circle plus an orientation-flip coset (`MassCovarianceForcing`, `CovarianceGroupFull`). The remaining generator-level gate is now closed: covariance of  $H(k, z) = k\Gamma + B_z$  is exactly the diagonal chiral-phase circle at fixed momentum, while the antidiagonal coset survives only together with parity  $k \mapsto -k$ . These same two branches now act exactly on the ordered two-channel finite step  $U(k, z) = T(k)C(z)$ : the diagonal branch sends  $z \mapsto uz$  at fixed momentum, while the antidiagonal branch sends  $(k, z) \mapsto (-k, u\bar{z})$ . This is `NEW KERNEL (DiscreteWalkMassCovariance)`. Exhaustion of every covariance of that discrete step, and its extension to the full ordered  $3 + 1$  regulator, remain open. A covariance transforms the field parameter; it is not literally a commutant.

7. **Add a scale-bearing selection mechanism and derive couplings.** The generic radial theorem now proves that no positive fixed finite homogeneous action of positive natural degree has a nonzero stationary scale; degree zero is flat. Therefore a nonzero solution must display a refining RG family, competing homogeneity, constraint, ensemble scale, or other dimensionful input and quantify which observed scale remains supplied. Interaction strengths and gauge representations remain open.

The finite construction encodes spacetime information algebraically through a fixed Pauli map and null direction grading. Emergence of a dynamical continuum geometry remains conjectural. We therefore do not claim that the paper starts without spacetime structure; it starts without primitive timelike motion or a primitive mass term inside the finite model.

## 11 Conclusion

The result readers should retain is compact. Null spinors open an exterior area. That area is the positive rest gap, its square is the momentum determinant, and its complex orientation is the off-diagonal part of an odd Hermitian mass operator. The collinear null boundary is exactly the gap-closing locus.

The operator does more than rename a scalar. It produces the relativistic Dirac square, exponentiates to the exact unitary turn, and fixes the finite corner ratio, including the directed  $z/\bar{z}$  factors in the exact history sum. The associated split-step walk converges uniformly on bounded  $1+1$  momentum boxes at an explicit  $O(1/n)$  rate, and its transfer-matrix history representation has a two-weight recursion that scales to the checkerboard corner ratio  $-i \tan(a\mu)$ . The finite local  $3+1$  extension accepts arbitrary position-dependent Pluecker profiles, preserves norm exactly, has an explicit inverse and causal cone, and turns local phase variation into endpoint links. Its ordered symbol carries the complex phase at the same uniform compact-box rate, the three-torus Plancherel identity fixes the exact  $N^{-3}$  wave-packet normalization, and compactly supported momentum-space data converge in  $L^2$  at every fixed time. The site field supplies its own patched link data with an exact integer winding, generated by primitive spinors. A common phase is invisible to the one-particle rest spectrum, but transported relative phase changes the exact two-site free spectrum through  $\Delta = z_R - (\bar{w})^2 z_L$ ; the supplied pair-kick sector provides a second, two-particle interference readout. These statements form one finite chain from null geometry to rest energy and propagation. The same audit proves where that chain presently stops: the cubic regulator has three exact zero-quasienergy corner aliases, its massive body-center step retains exact 0 and  $\pi$  quasienergies for every mass angle, and for every intermediate mass angle those body-center-type points are provably the only zero- or  $\pi$ -quasienergy crossings. The degree-one stationary-amplitude no-go excludes the simplest attempted repair, and its factorized form proves that every onsite mass coin leaves all three even-parity aliases intact. A Wilson term removes the non-origin corners and opens a uniform massive gap at the local-Hamiltonian level; converting that success into a strictly finite-range discrete-time regulator remains open. Exact temporal blocking further shows that the mass subgroup closes while the full split family generates a larger ordered monoid, and the homogeneous-action no-go proves that a nonzero absolute mass needs an additional scale-bearing input.

What remains is equally definite: a changing-lattice position-space PDE limit, a local strictly finite-range unitary regulator outside the degree-one stationary-amplitude no-go that matches the Wilson Hamiltonian's spectral success without sacrificing the derived coin, a blocked comparison with the tetrahedral construction, a patched defect/index theorem, and a second-quantized interacting extension. Success would extend a machine-verified finite rest-gap construction into a broader discrete relativistic dynamics. Failure at any named step would locate the boundary of the idea rather than dissolve it into analogy.

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